FACULTY OF ENGINEERING • KHON KAEN UNIVERSITY

Fundamental Electrical Engineering (FunE)

Final Exam Master Flashcards & Solved Problems Booklet (73 Cards)

COURSE & INSTRUCTOR
Assoc. Prof. Nararat Ruangchaijatupon

COVERAGE
Chapters 6, 7, 8, 9, 10 & Semiconductors (100% Complete)

CONTENT INCLUSIONS
73 Cards • 35 Authentic Diagrams • Line-by-Line Math Derivations

PART 1

Chapter 6: Three-Phase Circuits

Balanced & Unbalanced Y-Y, Y-Δ, Complex Power, Neutral Shift, 2-Wattmeter Method (10 Flashcards)

CH06-01 | Chapter 6 | Level 1 (Core Concept) | 📌 Lecture Slides & Formula Sheet

Star (Y) vs Delta (Δ) Voltage & Current Relations

📌 **PROBLEM / QUESTION:**

In a balanced 3-phase positive-sequence (abc) system, state the exact relationships between line quantities (V_L, I_L) and phase quantities (V_p, I_p) for both Y-connection and Δ-connection.

📐 **KEY FORMULAS & DEFINITIONS:**

Y: $V_L = \sqrt{3}V_p\angle +30^\circ,\ I_L = I_p$ Δ: $V_L = V_p,\ I_L = \sqrt{3}I_p\angle -30^\circ$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Star (Y) Connection:
- Line Voltage leads Phase Voltage by 30°: $V_{ab} = \sqrt{3}V_{an}\angle +30^\circ$
- Line Current equals Phase Current: $I_L = I_p$

2. Delta (Δ) Connection:
- Line Voltage equals Phase Voltage: $V_L = V_p$
- Line Current lags Phase Current by 30°: $I_a = \sqrt{3}I_{ab}\angle -30^\circ$

⚠️ **EXAM PITFALL & CAUTION:**

In negative sequence (acb), line voltage leads by -30° (or lags by 30°). Remember: in Y, currents are identical; in Δ, voltages are identical.

CH06-02 | Chapter 6 | Level 1 (Core Concept) | 📌 HW Ch.6 Problem 1 & 5

Delta to Wye (Δ ↔ Y) Impedance Conversion

📌 **PROBLEM / QUESTION:**

If a balanced 3ϕ load is connected in Δ with impedance $Z_\Delta = 60 + j30\ \Omega$, what is the equivalent impedance per phase when converted to a Wye (Y) connection (Z_Y)?

📐 **KEY FORMULAS & DEFINITIONS:**

$$Z_Y = \frac{Z_\Delta}{3} \quad \Leftrightarrow \quad Z_\Delta = 3Z_Y$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

$$Z_Y = \frac{Z_\Delta}{3} = \frac{60 + j30}{3} = 20 + j10\ \Omega$$

Conversely, converting from Wye to Delta: $Z_\Delta = 3Z_Y$.

⚠️ **EXAM PITFALL & CAUTION:**

Multiplying by 3 instead of dividing is a classic mistake! Delta load draws 3 times more current than Wye load of the same impedance, so equivalent Wye impedance must be 3 times smaller.

CH06-03 | Chapter 6 | Level 1 (Core Concept) | 📌 Formula Sheet & Lecture Slides

Three-Phase Power Formulas & Complex Power

📌 **PROBLEM / QUESTION:**

State the formulas for balanced 3ϕ Real Power ($P_{3\phi}$), Reactive Power ($Q_{3\phi}$), Apparent Power ($|S_{3\phi}|$), and Complex Power ($S_{3\phi}$) in terms of line values (V_L, I_L).

📐 **KEY FORMULAS & DEFINITIONS:**

$$P_{3\phi} = \sqrt{3}V_L I_L \cos\theta \quad [\text{W}] \quad Q_{3\phi} = \sqrt{3}V_L I_L \sin\theta \quad [\text{VAR}] \quad |S_{3\phi}| = \sqrt{3}V_L I_L \quad [\text{VA}] \quad S_{3\phi} = \sqrt{3}V_L I_L^* = P + jQ$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

When using Line values (V_L, I_L), always include the factor $\sqrt{3}$.
When using Phase values (V_p, I_p), use factor 3:
- $P = 3V_p I_p \cos\theta$
- $Q = 3V_p I_p \sin\theta$
- $S = 3V_p I_p^*$

where $\theta = \angle Z_p$ is the load phase impedance angle, and $PF = \cos\theta$.

⚠️ **EXAM PITFALL & CAUTION:**

Never mix $\sqrt{3}$ with phase quantities! It is either $\sqrt{3}V_L I_L$ or $3V_p I_p$.

CH06-04 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 1 (Page 1)

Balanced 4-Wire Y-Y System Full Calculation

PROBLEM / QUESTION:

In the balanced Y-Y circuit shown below, source phase voltages are $V_{an} = 220\angle 0^\circ$ V, $V_{bn} = 220\angle -120^\circ$ V, $V_{cn} = 220\angle -240^\circ$ V. Line impedance $Z_{\text{line}} = 10 + j5\ \Omega$. Balanced load impedance $Z_L = 20 + j50\ \Omega$. Find: (1) Line voltages V_{ab}, V_{bc}, V_{ca}, (2) Line currents I_{aA}, I_{bB}, I_{cC}, (3) Neutral current I_{Nn}, (4) Total 3ϕ complex power S.

active power and power factor.

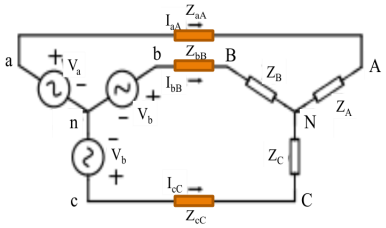


Fig. 1 Y to Y circuit with line impedance and balanced load.

รูปประกอบ: Homework 6 Figure 1: Balanced Y-Y System

KEY FORMULAS & DEFINITIONS:

$$V_{ab} = \sqrt{3}V_{an}\angle +30^\circ \quad I_{aA} = \frac{V_{an}}{Z_{\text{line}} + Z_L} \quad I_{Nn} = 0 \quad S_{3\phi} = 3V_{an}I_{aA}^*$$

STEP-BY-STEP MODEL SOLUTION:

1. Line Voltages:

- $V_{ab} = \sqrt{3}(220)\angle(0^\circ + 30^\circ) = 381.05\angle 30^\circ$ V
- $V_{bc} = 381.05\angle -90^\circ$ V
- $V_{ca} = 381.05\angle -210^\circ$ V

2. Per-phase total impedance:

$$Z_{\text{total}} = (10 + j5) + (20 + j50) = 30 + j55\ \Omega = 62.65\angle 61.39^\circ\ \Omega$$

3. Line Currents:

- $I_{aA} = \frac{220\angle 0^\circ}{62.65\angle 61.39^\circ} = 3.51\angle -61.39^\circ$ A
- $I_{bB} = 3.51\angle -181.39^\circ$ A
- $I_{cC} = 3.51\angle 58.61^\circ$ A

4. Neutral Current:

In a balanced system, $I_{Nn} = 0$ A, $V_{Nn} = 0$ V.

5. Total 3ϕ Power:

$$S = 3 \times 220 \times 3.51\angle +61.39^\circ = 2,316.6\angle 61.39^\circ\ \text{VA} = 1,109\ \text{W} + j2,034\ \text{VAR}$$

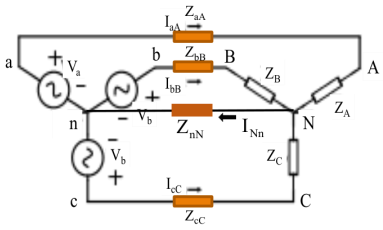
EXAM PITFALL & CAUTION:

Always remember to conjugate the current (I^*) when calculating complex power: $S = VI^*$. The angle of S equals the angle of Z.

Unbalanced 4-Wire Y-Y System with Neutral Wire

PROBLEM / QUESTION:

In the 4-wire Y-Y system shown below, the source is balanced $220\ \text{V}$ rms, line impedance $Z_{\text{line}} = 10 + j5\ \Omega$, neutral line impedance $Z_{nN} = 10 + j5\ \Omega$, and the load is UNBALANCED: $Z_A = 20 + j50\ \Omega$, $Z_B = 10 + j50\ \Omega$, $Z_C = 20 + j20\ \Omega$. How do you solve for the Neutral Shift Voltage (V_{Nn}) and line currents?



ig. 2 Y to Y circuit with line impedance and unbalanced load.

ed from: https://en.wikipedia.org/wiki/Three-phase_electric_power

รูปประกอบ: Homework 6 Figure 2: Unbalanced Y-Y 4-Wire with Neutral Line

KEY FORMULAS & DEFINITIONS:

$$V_{Nn} = \frac{\frac{V_{an}}{Z'_a} + \frac{V_{bn}}{Z'_b} + \frac{V_{cn}}{Z'_c}}{\frac{1}{Z'_a} + \frac{1}{Z'_b} + \frac{1}{Z'_c} + \frac{1}{Z_{nN}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Millman's / Node Voltage at Neutral node N:
Let node n be the reference ($0\ \text{V}$).
Total phase branch impedances:
- $Z'_a = (10 + j5) + (20 + j50) = 30 + j55\ \Omega$
- $Z'_b = (10 + j5) + (10 + j50) = 20 + j55\ \Omega$
- $Z'_c = (10 + j5) + (20 + j20) = 30 + j25\ \Omega$
- $Z_{nN} = 10 + j5\ \Omega$

2. Solve for V_{Nn} using KCL at node N:

$$\frac{V_N - V_{an}}{Z'_a} + \frac{V_N - V_{bn}}{Z'_b} + \frac{V_N - V_{cn}}{Z'_c} + \frac{V_N}{Z_{nN}} = 0$$

3. Calculate Line Currents:
- $I_{aA} = \frac{V_{an} - V_{Nn}}{Z'_a}$
- $I_{bB} = \frac{V_{bn} - V_{Nn}}{Z'_b}$
- $I_{cC} = \frac{V_{cn} - V_{Nn}}{Z'_c}$
- $I_{Nn} = \frac{V_{Nn}}{Z_{nN}} = I_{aA} + I_{bB} + I_{cC}$

EXAM PITFALL & CAUTION:

Do NOT assume $V_{Nn} = 0$! In an unbalanced system, current flows through the neutral wire, causing a non-zero neutral shift voltage $V_{Nn} \neq 0$.

CH06-06 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 3 (Page 2)

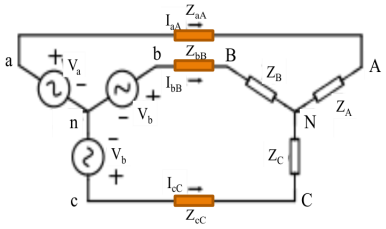
Unbalanced 3-Wire Y-Y System (NO Neutral Wire)

PROBLEM / QUESTION:

In the 3-wire Y-Y system shown below (NO neutral wire), the load is unbalanced ($Z_A \neq Z_B \neq Z_C$). How is the neutral shift voltage V_{Nn} calculated, and what happens to the neutral impedance term in the Millman formula?

ents (I_{aA}, I_{bB}, I_{cC}), I_{Nn}, phase voltages (V_{AN}, V_{BN}, V_{CN}) of loads, 3ϕ complex

al power, 3ϕ reactive power and power factor.



รูปประกอบ: Homework 6 Figure 3: Unbalanced Y-Y 3-Wire System

KEY FORMULAS & DEFINITIONS:

$$V_{Nn} = \frac{\frac{V_{an}}{Z'_a} + \frac{V_{bn}}{Z'_b} + \frac{V_{cn}}{Z'_c}}{\frac{1}{Z'_a} + \frac{1}{Z'_b} + \frac{1}{Z'_c}}$$

$$I_{aA} + I_{bB} + I_{cC} = 0$$

STEP-BY-STEP MODEL SOLUTION:

1. Floating Neutral Point:
Because there is no neutral conductor, $Z_{nN} = \infty$, meaning the admittance term $\frac{1}{Z_{nN}} = 0$.

2. Node Voltage equation at N:

$$V_{Nn} = \frac{V_{an}Y'_a + V_{bn}Y'_b + V_{cn}Y'_c}{Y'_a + Y'_b + Y'_c}$$

where $Y'_x = \frac{1}{Z'_x}$.

3. Zero Neutral Current:
Without a neutral path, $I_{Nn} = 0$ strictly holds, which means:

$$I_{aA} + I_{bB} + I_{cC} = 0$$

EXAM PITFALL & CAUTION:

Even though $I_{Nn} = 0$ (no physical wire), V_{Nn} is NOT zero! The neutral point floats and shifts severely, subjecting some phases to dangerous overvoltages.

CH06-07 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 4 (12 HP Y-Motor)

3-Phase Motor Input Power, Efficiency & Line Current

PROBLEM / QUESTION:

A 3ϕ, 12-hp induction motor is connected in Y to a balanced $380\ \text{V}, 50\ \text{Hz}$ supply. The motor has efficiency $\eta = 90\%$ and $PF = 0.85$ lagging. Calculate: (1) Input electrical power P_{in}, (2) Line current I_L, (3) Total complex power S, (4) Equivalent impedance per phase Z_Y.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{out}} = 12 \times 746\ \text{W} = 8,952\ \text{W}$$

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta}$$

$$I_L = \frac{P_{\text{in}}}{\sqrt{3} V_L \cdot PF}$$

$$S = \frac{P_{\text{in}}}{PF} \angle \arccos(PF)$$

STEP-BY-STEP MODEL SOLUTION:

1. Convert output power to Watts:

$$P_{\text{out}} = 12 \times 746 = 8,952\ \text{W}$$

2. Input Electrical Real Power:

$$P_{\text{in}} = \frac{8,952}{0.90} = 9,946.67\ \text{W} \approx 9.95\ \text{kW}$$

3. Line Current (I_L):

$$I_L = \frac{9,946.67}{\sqrt{3} \times 380 \times 0.85} = \frac{9,946.67}{559.45} = 17.78\ \text{A}$$

4. Complex Power (S):

$$\theta = \arccos(0.85) = 31.79^\circ$$

$$|S| = \frac{9.946}{0.85} = 11.70\ \text{kVA}$$

$$S = 11.70 \angle 31.79^\circ\ \text{kVA} = 9.95\ \text{kW} + j6.16\ \text{kVAR}$$

5. Per-Phase Impedance (Z_Y):

$$V_p = \frac{380}{\sqrt{3}} = 219.39\ \text{V}$$

$$Z_Y = \frac{V_p}{I_L} = \frac{219.39}{17.78} \angle 31.79^\circ = 12.34 \angle 31.79^\circ\ \Omega = 10.49 + j6.50\ \Omega$$

($R = 10.49\ \Omega$, $L = \frac{6.50}{2\pi \times 50} = 20.7\ \text{mH}$)

EXAM PITFALL & CAUTION:

Remember: $1\ \text{hp} = 746\ \text{W}$. Horsepower is ALWAYS mechanical output power at the shaft (P_{out}), never electrical input power (P_{in})!

CH06-08 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 5 (10 HP Δ-Motor)

Delta-Connected 3-Phase Motor Analysis

PROBLEM / QUESTION:

A $3\phi, 10\text{-hp}$ induction motor is connected in Δ to a $380\ \text{V}, 50\ \text{Hz}$ supply. $\eta = 85\%, PF = 0.80$ lagging. Find: (1) Input power P_{in}, (2) Line current I_L, (3) Phase winding current I_Δ, (4) Phase impedance Z_Δ.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{in}} = \frac{10 \times 746}{0.85}$$
$$I_L = \frac{P_{\text{in}}}{\sqrt{3} V_L PF}$$
$$I_\Delta = \frac{I_L}{\sqrt{3}}$$
$$Z_\Delta = \frac{V_L}{I_\Delta}$$

STEP-BY-STEP MODEL SOLUTION:

1. Input Power:

$$P_{\text{in}} = \frac{7,460\ \text{W}}{0.85} = 8,776.47\ \text{W} = 8.78\ \text{kW}$$

2. Line Current:

$$I_L = \frac{8,776.47}{\sqrt{3} \times 380 \times 0.80} = 16.67\ \text{A}$$

3. Delta Phase Current:
In Δ, line current divides: $I_\Delta = \frac{I_L}{\sqrt{3}} = \frac{16.67}{1.732} = 9.62\ \text{A}$

4. Impedance per phase (Z_Δ):
In Δ, $V_p = V_L = 380\ \text{V}$.

$$|Z_\Delta| = \frac{V_p}{I_\Delta} = \frac{380}{9.62} = 39.50\ \Omega$$

$$\theta = \arccos(0.8) = 36.87^\circ$$

$$Z_\Delta = 39.50 \angle 36.87^\circ\ \Omega = 31.60 + j23.70\ \Omega$$

EXAM PITFALL & CAUTION:

In a Delta load, $V_p = V_L = 380\ \text{V}$, but $I_p = I_L/\sqrt{3}$. Do not divide voltage by $\sqrt{3}$ for a Delta load!

Two-Wattmeter Method for 3-Phase Power Measurement

📌 **PROBLEM / QUESTION:**

In the two-wattmeter method measuring a balanced 3-phase load, state the equations for total real power (P_{total}), total reactive power (Q_{total}), and the load power factor angle (θ) in terms of wattmeter readings W_1 and W_2.

📐 **KEY FORMULAS & DEFINITIONS:**

$$P_{\text{total}} = W_1 + W_2$$
$$Q_{\text{total}} = \sqrt{3}(W_1 - W_2)$$
$$\tan\theta = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Total Active Power is simply the algebraic sum:

$$P = W_1 + W_2$$

2. Total Reactive Power is proportional to the difference:

$$Q = \sqrt{3}(W_1 - W_2)$$

3. Phase angle of the load:

$$\theta = \arctan\left(\sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}\right)$$

$$PF = \cos\theta$$

Special Cases:
- If $PF = 1.0$: $W_1 = W_2$
- If $PF = 0.5$: One wattmeter reads zero ($W_2 = 0$)
- If $PF < 0.5$: One wattmeter reads NEGATIVE!

⚠️ **EXAM PITFALL & CAUTION:**

When $PF < 0.5$, one wattmeter reads negative! On an analog meter, reverse the current coil connections to read the value, then treat that reading as a negative number in $W_1 + W_2$.

CH06-10 | Chapter 6 | Level 2 (Exam / Calc) | Lecture Slides & Final Review

Power Factor Correction with Delta Capacitors

PROBLEM / QUESTION:

A balanced 3ϕ inductive load of $30\ \text{kW}$ operates at $380\ \text{V}, 50\ \text{Hz}$ with $PF_1 = 0.65$ lagging. What total reactive power Q_C must a capacitor bank supply to improve the power factor to $PF_2 = 0.95$ lagging, and what is the capacitance per phase if connected in Delta (C_Δ)?

KEY FORMULAS & DEFINITIONS:

$$Q_C = P(\tan\theta_1 - \tan\theta_2)$$

$$C_\Delta = \frac{Q_{C,\text{phase}}}{\omega V_L^2} = \frac{Q_C/3}{2\pi f V_L^2}$$

STEP-BY-STEP MODEL SOLUTION:

1. Find power factor angles:
- $\theta_1 = \arccos(0.65) = 49.46^\circ \Rightarrow \tan\theta_1 = 1.169$
- $\theta_2 = \arccos(0.95) = 18.19^\circ \Rightarrow \tan\theta_2 = 0.329$

2. Required 3-phase Reactive Power from Capacitors:

$$Q_C = 30\ \text{kW} \times (1.169 - 0.329) = 30 \times 0.840 = 25.20\ \text{kVAR}$$

3. Delta connection per-phase capacitance:

Each branch sees full line voltage $V_L = 380\ \text{V}$ and supplies $Q_{C1} = \frac{25{,}200}{3} = 8{,}400\ \text{VAR}$:

$$C_\Delta = \frac{8{,}400}{2\pi(50)(380)^2} = \frac{8{,}400}{314.16 \times 144{,}400} = 1.85 \times 10^{-4}\ \text{F} = 185\ \mu\text{F}$$

EXAM PITFALL & CAUTION:

Capacitors connected in Delta require 3 times LESS capacitance than Wye-connected capacitors ($C_\Delta = C_Y/3$) for the same reactive power, saving massive physical space and cost!

PART 2

Chapter 7: Electrical Machines (DC & AC)

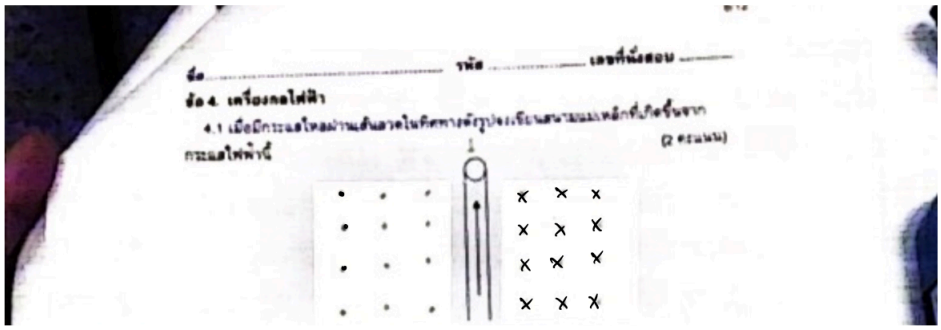
Right-Hand Rule, Lorentz Force, DC Shunt/Series Motors & Generators, Induction & Synchronous (15 Flashcards)

CH07-01 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.1 (Page 28 - 2 Points)

Right-Hand Grip Rule for Magnetic Field around a Conductor

📌 **PROBLEM / QUESTION:**

Current flows upward along a straight vertical wire as shown below. Using the right-hand grip rule, sketch/state the direction of the magnetic field ($\vec{B}$) on both sides of the wire.



รูปประกอบ: Exam Problem 4.1: Right-Hand Grip Rule

📐 **KEY FORMULAS & DEFINITIONS:**

Thumb = Current (I) Fingers curl = Magnetic field ($\vec{B}$)

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Align the right thumb with current flow pointing UPWARD.
2. Curled fingers point:
- Out of the page on the LEFT side (represented by dots •)
- Into the page on the RIGHT side (represented by crosses ×)

⚠️ **EXAM PITFALL & CAUTION:**

Always use the RIGHT hand! Left-hand grip reverses all field directions.

CH07-02 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.2 (Page 28 - 3 Points)

DC Motor Magnetic Force on Wire (Lorentz Force Rule)

📌 PROBLEM / QUESTION:

In the DC motor setup shown below, the upper permanent magnet pole is North (N) and the lower is South (S). Current is applied to the loop. State: (1) Current direction, (2) Direction of magnetic field $\vec{B}$, (3) Direction of resulting force $\vec{F}$ on each coil side.



รูปประกอบ: Exam Problem 4.2: DC Motor Magnetic Field & Force (Lorentz)

📐 KEY FORMULAS & DEFINITIONS:

$$\vec{F} = I(\vec{L} \times \vec{B})$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Magnetic field $\vec{B}$ points from North to South (DOWNWARD).
2. Current flows from (+) terminal through the coil to (-) terminal.
3. By Fleming's Left-Hand Rule (or $\vec{L} \times \vec{B}$):
- The two opposite coil sides experience opposing forces forming a clockwise or counterclockwise torque that rotates the rotor shaft.

⚠️ EXAM PITFALL & CAUTION:

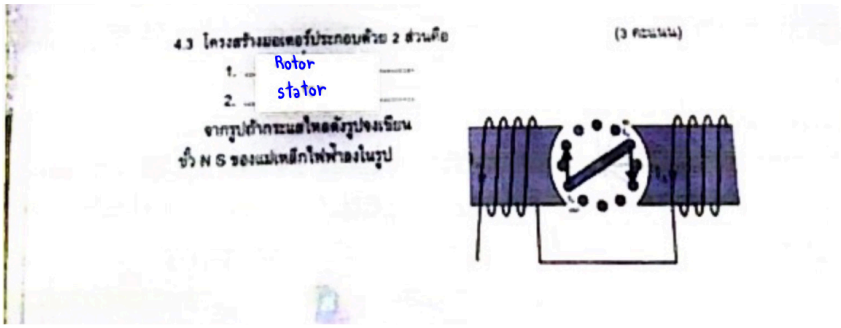
For motors, use Fleming's LEFT-hand rule. For generators, use Fleming's RIGHT-hand rule.

CH07-03 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.3 (Page 28 - 3 Points)

DC Machine Stator vs Rotor & Electromagnet Pole Polarities

📌 **PROBLEM / QUESTION:**

What are the two primary structural parts of an electric motor? For the electromagnet core shown below, identify the magnetic poles (N and S) established by the coil current.



รูปประกอบ: Exam Problem 4.3: Motor Magnetic Poles (N & S)

📐 **KEY FORMULAS & DEFINITIONS:**

Right-hand solenoid rule: Fingers = Current curl, Thumb = North Pole (N)

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. The two essential parts are:
- Stator (ส่วนที่อยู่กับที่ - stationary field frame)
- Rotor / Armature (ส่วนที่หมุนเคลื่อนที่ - rotating part)

2. Solenoid Pole Identification:
Using the right-hand coil rule, curl four fingers in the direction current wraps around the iron pole pieces. The thumb points toward the North pole (where magnetic flux emerges).

⚠️ **EXAM PITFALL & CAUTION:**

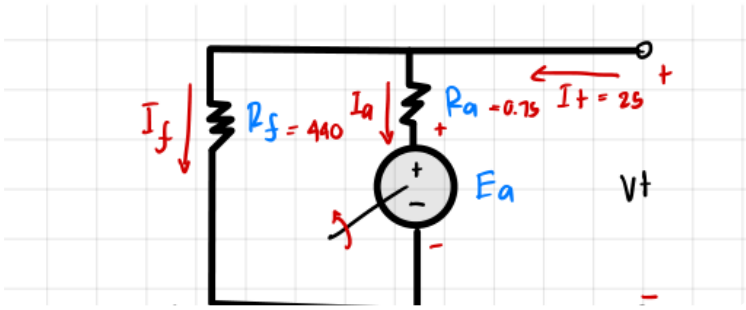
Do not confuse the armature core with the stator frame. Stator provides the stationary B-field; rotor carries the torque-producing coils.

CH07-04 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.4 (Page 27 - 8 Points)

3-Phase Induction Motor & Squirrel-Cage Rotor Operation

📌 **PROBLEM / QUESTION:**

Referring to the 3-phase induction motor diagram below: (1) What is a Squirrel-Cage Rotor and how does it work? (2) Explain the operating principle of a 3-phase induction motor. (3) How do you reverse its direction of rotation?



รูปประกอบ: Homework 7 Problem 4: DC Shunt Motor Equivalent Circuit

📐 **KEY FORMULAS & DEFINITIONS:**

$$n_s = \frac{120f}{P}$$

$$s = \frac{n_s - n_r}{n_s}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Squirrel-Cage Rotor (โรเตอร์แบบกรงกระรอก):
Consists of conducting aluminum/copper bars embedded in laminated iron core, short-circuited at both ends by end-rings. When the rotating magnetic field sweeps across the bars, it induces currents producing torque.

2. Operating Principle:
3-phase stator currents produce a Constant-Magnitude Rotating Magnetic Field (RMF) spinning at synchronous speed $n_s = 120f/P$. By Faraday-Lenz law, this field cuts rotor bars, inducing voltage and rotor currents. The interaction generates Lorentz torque, pulling the rotor in the direction of the rotating field.

3. Direction Reversal:
Swap ANY TWO supply phase lines (e.g., swap A and B, keeping C fixed). This reverses the phase sequence and rotates the RMF in the opposite direction.

⚠️ **EXAM PITFALL & CAUTION:**

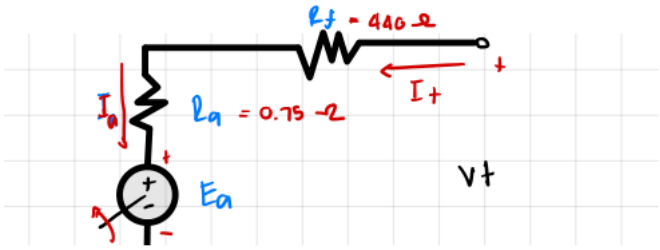
The rotor can NEVER reach synchronous speed ($n_r < n_s$). If $n_r = n_s$, relative velocity is zero, no voltage is induced, and torque drops to zero!

CH07-05 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.5 (Page 26 - 4 Points)

DC Generator Principle & Commutator Rectification

📌 **PROBLEM / QUESTION:**

What machine is shown in the diagram below and how does it work? Explain how the split-ring commutator converts induced AC into pulsating DC at the output terminals.



รูปประกอบ: Homework 7 Problem 5: DC Series Motor Equivalent Circuit

📐 **KEY FORMULAS & DEFINITIONS:**

$$e = B\ell v \sin\theta$$

$$E_a = k_a \phi \omega_m$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Identification:
Single-phase DC Generator with split-ring commutator.

2. Principle of Operation:
Mechanical power drives the coil rotating in the magnetic field between permanent magnet poles. By Faraday's Law, an alternating electromotive force ($e = B\ell v$) is induced inside the rotating coil.

3. Commutator Action:
The split-ring commutator mechanically reverses coil connections to the external stationary carbon brushes every half revolution, rectifying the internal alternating AC into unidirectional pulsating DC.

⚠️ **EXAM PITFALL & CAUTION:**

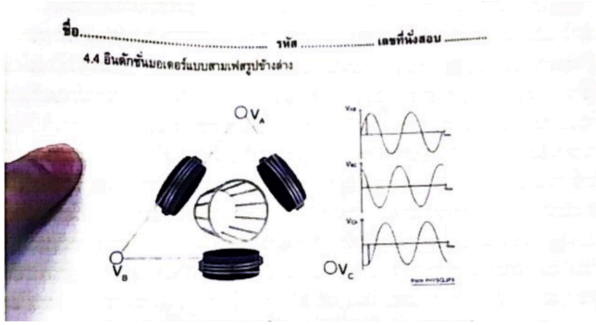
Inside ANY rotating DC armature, the induced voltage is ALWAYS AC! The commutator is what converts internal AC into external DC.

CH07-06 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.7 Problem 4 & Talui Homework Page 1

DC Shunt Motor Equivalent Circuit, Speed & Torque

PROBLEM / QUESTION:

A 2-pole, $2.5\text{-hp}, 100\ \text{VDC}$ shunt motor has design constant $k_a = 153$. At full load, $I_t = 25.0\ \text{A}, R_a = 0.75\ \Omega, \phi = 10.0\ \text{mWb}, R_f = 440\ \Omega$. Draw the equivalent circuit, and calculate: (1) Back EMF E_a, (2) Rotational speed n [rpm], (3) Developed torque τ_d [N·m].



รูปประกอบ: Exam Problem 4.4: 3-Phase Induction Motor (Squirrel Cage)

KEY FORMULAS & DEFINITIONS:

$$I_f = \frac{V_t}{R_f}, \quad I_a = I_t - I_f$$
$$E_a = V_t - I_a R_a = k_a \phi \omega_m$$
$$\omega_m = \frac{2\pi n}{60}, \quad \tau_d = k_a \phi I_a$$

STEP-BY-STEP MODEL SOLUTION:

1. Field current & Armature current:

$$I_f = \frac{V_t}{R_f} = \frac{100}{440} = 0.23\ \text{A}$$

$$I_a = I_t - I_f = 25.0 - 0.23 = 24.77\ \text{A}$$

2. Back EMF (E_a):

$$E_a = V_t - I_a R_a = 100 - (24.77 \times 0.75) = 100 - 18.58 = 81.42\ \text{V}$$

3. Speed (n):

$$E_a = k_a \phi \omega_m \Rightarrow 81.42 = 153 \times (10 \times 10^{-3}) \times \omega_m$$

$$\omega_m = \frac{81.42}{1.53} = 53.22\ \text{rad/s}$$

$$n = \frac{60 \times \omega_m}{2\pi} = \frac{60 \times 53.22}{2\pi} = 508.17\ \text{rpm}$$

4. Developed Torque (τ_d):

$$\tau_d = k_a \phi I_a = 153 \times (10 \times 10^{-3}) \times 24.77 = 37.90\ \text{N} \cdot \text{m}$$

EXAM PITFALL & CAUTION:

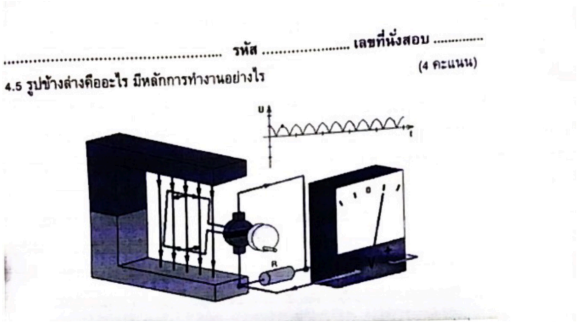
In a Shunt motor, $I_t = I_a + I_f$. Do NOT use total line current I_t in $V_t - IR_a$; only I_a passes through the armature resistance!

CH07-07 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.7 Problem 5 & Talui Homework Page 2

DC Series Motor Circuit & Terminal Voltage Calculation

PROBLEM / QUESTION:

A 2-pole, 2.5-hp DC Series motor has $k_a = 153, R_a = 0.75\ \Omega, R_f = 440\ \Omega, \phi = 10.0\ \text{mWb}$. Find: (a) Terminal voltage V_{t1} giving the SAME SPEED ($n = 508.17$ rpm) as Problem 4, (b) Terminal voltage V_{t2} giving the SAME TORQUE ($\tau_d = 37.9\ \text{N} \cdot \text{m}$) as Problem 4.



รูปประกอบ: Exam Problem 4.5: DC Generator Principle & Split-Ring Commutator

KEY FORMULAS & DEFINITIONS:

Series connection: $I_t = I_a = I_f$

$$V_t = E_a + I_a(R_a + R_f)$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Same Speed ($n = 508.17$ rpm):

$E_a = k_a \phi \omega_m = 81.42\ \text{V}$.

From power $P_e = E_a I_a = 1,865\ \text{W} \Rightarrow I_a = \frac{1,865}{81.42} = 22.91\ \text{A}$.

$$V_{t1} = E_a + I_a(R_a + R_f) = 81.42 + 22.91(0.75 + 440) = 81.42 + 10,097.6 = 10.18\ \text{kV}$$

2. Part (b) Same Torque ($\tau_d = 37.9\ \text{N} \cdot \text{m}$):

From $\tau_d = k_a \phi I_a \Rightarrow I_a = 24.77\ \text{A}$.

$P_e = E_a I_a = 1,865\ \text{W} \Rightarrow E_a = \frac{1,865}{24.77} = 75.30\ \text{V}$.

$$V_{t2} = E_a + I_a(R_a + R_f) = 75.30 + 24.77(440.75) = 11.00\ \text{kV}$$

EXAM PITFALL & CAUTION:

In a Series motor, the field winding is in series with the armature ($I_a = I_f = I_t$). The total circuit resistance is $R_a + R_f$.

CH07-08 | Chapter 7 | Level 2 (Exam / Calc) | FunE-final.pdf Page 5 (Example 3)

DC Shunt Motor Speed Control via Armature Resistance

PROBLEM / QUESTION:

A $220\ \text{V}$ DC shunt motor has $R_a = 0.5\ \Omega, R_f = 110\ \Omega$. At full load, $I_t = 42\ \text{A}$ and speed $n_1 = 1,200\ \text{rpm}$. If an external resistor $R_{\text{ext}} = 1.0\ \Omega$ is inserted in series with the armature while torque remains constant, calculate the new motor speed n_2.

KEY FORMULAS & DEFINITIONS:

$$E_{a1} = V_t - I_{a1}R_a$$
$$E_{a2} = V_t - I_{a2}(R_a + R_{\text{ext}})$$
$$\frac{n_2}{n_1} = \frac{E_{a2}}{E_{a1}} \quad (\text{when } \phi \text{ is constant})$$

STEP-BY-STEP MODEL SOLUTION:

1. Initial condition:

$$I_f = \frac{220}{110} = 2.0\ \text{A}$$

$$I_{a1} = 42 - 2 = 40.0\ \text{A}$$

$$E_{a1} = 220 - (40 \times 0.5) = 220 - 20 = 200\ \text{V}$$

2. New condition with $R_{\text{ext}} = 1.0\ \Omega$:
Constant torque and constant flux $\Rightarrow I_{a2} = I_{a1} = 40.0\ \text{A}$.

$$E_{a2} = 220 - 40(0.5 + 1.0) = 220 - 60 = 160\ \text{V}$$

3. Calculate new speed:

$$\frac{n_2}{1200} = \frac{160}{200} = 0.80 \Rightarrow n_2 = 1,200 \times 0.80 = 960\ \text{rpm}$$

EXAM PITFALL & CAUTION:

Adding armature resistance can ONLY reduce speed below base speed ($n < n_{\text{base}}$). To run above base speed, field weakening (reducing I_f) is required.

CH07-09 | Chapter 7 | Level 2 (Exam / Calc) | FunE-final.pdf Page 4 (Example 2)

DC Shunt Generator Voltage Regulation & KVL

PROBLEM / QUESTION:

A DC shunt generator driven at $1,800$ rpm produces no-load terminal voltage $V_t = 115$ V. When connected to a load, $I_a = 40$ A and terminal voltage drops to $V_t = 100$ V. Find: (1) Armature resistance R_a, (2) Voltage regulation ($VR\%$), (3) Counter-torque τ_d at full load if $\phi = 10$ mWb.

KEY FORMULAS & DEFINITIONS:

Generator KVL:

$$E_a = V_t + I_a R_a$$

$$VR\% = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

STEP-BY-STEP MODEL SOLUTION:

1. Armature Resistance (R_a):
At no load ($I_a = 0$), $E_a = V_{NL} = 115$ V.
At full load: $115 = 100 + (40 \times R_a) \Rightarrow R_a = \frac{15}{40} = 0.375\ \Omega$.

2. Voltage Regulation:

$$VR\% = \frac{115 - 100}{100} \times 100\% = 15.0\%$$

3. Counter Torque (τ_d):

$$\omega_m = \frac{2\pi(1,800)}{60} = 60\pi = 188.5 \text{ rad/s}$$

$$P_e = E_a I_a = 115 \times 40 = 4,600 \text{ W}$$

$$\tau_d = \frac{P_e}{\omega_m} = \frac{4,600}{188.5} = 24.40 \text{ N} \cdot \text{m}$$

EXAM PITFALL & CAUTION:

Generator KVL has a PLUS sign ($E_a = V_t + I_a R_a$), while Motor KVL has a MINUS sign ($E_a = V_t - I_a R_a$)! In a generator, internal generated EMF is higher than terminal voltage.

CH07-10 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Induction Motor Synchronous Speed, Rotor Speed & Slip

📌 **PROBLEM / QUESTION:**

A 4-pole, $50\ \text{Hz}$ three-phase induction motor runs at $1,440\ \text{rpm}$ at full load. Calculate: (1) Synchronous speed n_s, (2) Fractional slip s and percent slip $s\%$, (3) Rotor current frequency f_r, (4) Rotor speed at locked rotor ($s = 1$).

📐 **KEY FORMULAS & DEFINITIONS:**

$$n_s = \frac{120f}{P}$$
$$s = \frac{n_s - n_r}{n_s}$$
$$f_r = sf$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Synchronous Speed:

$$n_s = \frac{120 \times 50}{4} = 1,500\ \text{rpm}$$

2. Slip (s):

$$s = \frac{1,500 - 1,440}{1,500} = \frac{60}{1,500} = 0.04 = 4.0\%$$

3. Rotor Current Frequency (f_r):

$$f_r = s \times f = 0.04 \times 50\ \text{Hz} = 2.0\ \text{Hz}$$

4. At Locked-Rotor:

$n_r = 0$, slip $s = 1.0$, and rotor frequency equals line frequency ($f_r = 50\ \text{Hz}$).

⚠️ **EXAM PITFALL & CAUTION:**

Rotor electrical frequency is sf, NOT f! At normal running speed, slip is small ($2 - 5\%$), so rotor frequency is tiny ($1 - 2.5\ \text{Hz}$).

CH07-11 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Final Exam

Induction Motor Power Flow: Air-Gap, Copper Loss & Mech Power

PROBLEM / QUESTION:

State the universal proportional power relationship between Air-gap Power (P_{ag}), Rotor Copper Loss (P_{rcu}), and Converted Mechanical Power (P_{conv}) in an induction motor.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{ag}} : P_{\text{rcu}} : P_{\text{conv}} = 1 : s : (1-s)$$
$$P_{\text{rcu}} = sP_{\text{ag}}$$
$$P_{\text{conv}} = (1-s)P_{\text{ag}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Total air-gap power crosses the stator-rotor gap: P_{ag}.
2. Fraction s is dissipated as rotor I^2R heat: $P_{\text{rcu}} = sP_{\text{ag}}$.
3. Fraction $(1-s)$ is converted into mechanical power: $P_{\text{conv}} = (1-s)P_{\text{ag}}$.
4. Internal developed torque is: $\tau_d = \frac{P_{\text{ag}}}{\omega_s} = \frac{P_{\text{conv}}}{\omega_m}$.

EXAM PITFALL & CAUTION:

Rotor copper loss is DIRECTLY proportional to slip! High slip means high rotor heating and very poor electrical efficiency.

CH07-12 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Final Exam

Synchronous Motor Operation, V-Curve & Power Factor Correction

PROBLEM / QUESTION:

Why does a Synchronous Motor rotate at exactly synchronous speed with zero slip ($s = 0$)? How does varying its DC field excitation current (I_f) allow it to act as a Synchronous Condenser to correct facility power factor?

KEY FORMULAS & DEFINITIONS:

$$n_r = n_s = \frac{120f}{P}$$

Normal excitation: $PF = 1.0$ Over-excitation: Leading PF (supplies Q) Under-excitation: Lagging PF (draws Q)

STEP-BY-STEP MODEL SOLUTION:

1. Constant Synchronous Speed:
The rotor's DC-excited poles lock magnetically into synchronism with the stator's rotating magnetic field. As long as load does not exceed pull-out torque, it runs at precisely n_s.

2. V-Curve & Power Factor Control:
- Under-excited (I_f small): Motor draws lagging reactive power ($Q > 0$).
- Normal excitation: Motor operates at unity power factor ($PF = 1.0$), minimum stator current.
- Over-excited (I_f large): Motor draws leading reactive power (acts as a capacitor, supplying VARs to the grid to correct factory power factor).

EXAM PITFALL & CAUTION:

Synchronous motors are NOT inherently self-starting from a dead stop because the stator RMF moves past stationary rotor poles too fast for inertia to catch up. They require damper (amortisseur) windings to start as an induction motor first.

CH07-13 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Final Review

DC Motor Starting Current & Starting Resistor Design

PROBLEM / QUESTION:

Why is the starting current of a DC motor dangerously high (10 to 20 times rated current) if started directly on full voltage? How does an external starting resistor protect the motor?

KEY FORMULAS & DEFINITIONS:

At start ($n = 0$): $E_a = 0$

$$I_{\text{start}} = \frac{V_t}{R_a}$$

With starter:

$$I_{\text{start}} = \frac{V_t}{R_a + R_{\text{start}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Why Starting Current is Enormous:
At standstill ($n = 0$), back EMF $E_a = k_a \phi \omega_m = 0$ V. Because armature resistance R_a is designed to be tiny ($< 0.5\ \Omega$) for high efficiency:

$$I_{\text{start}} = \frac{V_t - 0}{R_a} = \frac{220\ \text{V}}{0.5\ \Omega} = 440\ \text{A!}$$

This burns the armature coils and melts commutator brushes.

2. Protection via Starting Resistor:
A multi-step starting resistance R_{start} is placed in series with the armature to limit starting current to $1.5 - 2.0 \times I_{\text{rated}}$. As the motor accelerates and builds up E_a, resistance steps are cut out.

EXAM PITFALL & CAUTION:

Never start a DC motor under full voltage without starting resistance (except tiny sub-fractional HP motors).

CH07-14 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Exam Pitfalls

Danger of Open Field Circuit in DC Shunt Motor

PROBLEM / QUESTION:

What happens if the shunt field circuit of a running DC shunt motor accidentally breaks or becomes open-circuited ($I_f = 0$)? Explain the physical mechanism.

KEY FORMULAS & DEFINITIONS:

$$n = \frac{V_t - I_a R_a}{k_a \phi}$$

As $\phi \to \phi_{\text{residual}} \approx 0 \Rightarrow n \to \infty$

STEP-BY-STEP MODEL SOLUTION:

1. Field current drops to zero ($I_f = 0$), so magnetic flux drops to residual magnetism ($\phi \to \phi_{\text{residual}} \approx 0$).
2. Back EMF collapses: $E_a \approx 0$, causing armature current $I_a = \frac{V_t - E_a}{R_a}$ to soar to destructive levels.
3. Because $n \propto \frac{1}{\phi}$, the motor accelerates uncontrollably to catastrophic runaway speed ($n \to \infty$), resulting in mechanical disintegration (throwing armature windings and bearings exploding).

EXAM PITFALL & CAUTION:

Never install a fuse or switch in the field winding of a DC shunt motor! An open field causes motor destruction.

CH07-15 | Chapter 7 | Level 1 (Core Concept) | Lecture Slides & Safety Rules

DC Series Motor Prohibition Against No-Load Operation

PROBLEM / QUESTION:

Why is it strictly forbidden to run a DC Series Motor without a mechanical load (or with a belt drive that could snap)?

KEY FORMULAS & DEFINITIONS:

Series motor: $\phi \propto I_a$

$$n \propto \frac{V_t - I_a(R_a + R_f)}{k_a \phi} \approx \frac{V_t}{k' I_a}$$

As $I_a \to 0 \Rightarrow n \to \infty$

STEP-BY-STEP MODEL SOLUTION:

1. In a series motor, field current equals armature current ($I_f = I_a$).
2. At no load, torque required is near zero, so armature current drops close to zero ($I_a \to 0$).
3. This causes magnetic flux to drop to residual levels ($\phi \to 0$).
4. Speed $n \propto \frac{1}{\phi} \propto \frac{1}{I_a}$ races towards dangerous infinity ($n \to \infty$), destroying the machine by centrifugal force.

EXAM PITFALL & CAUTION:

Series motors MUST ALWAYS be solidly coupled (gears/direct shaft) to their load, NEVER belt-driven!

PART 3

Chapter 8: Transformers

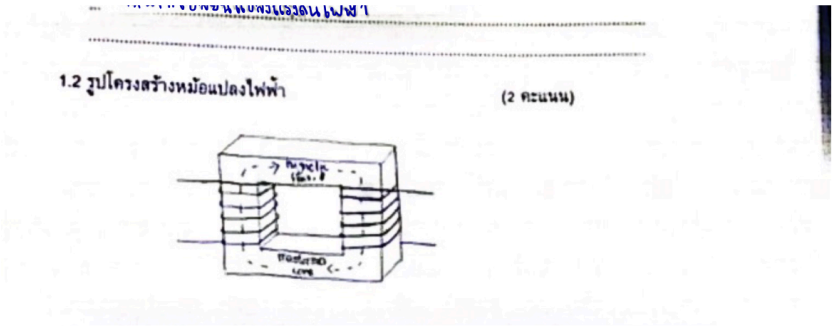
Mutual Induction, Induced EMF Formula, 25 kVA Exam Problem, Audio Matching, Efficiency & Auto-Transformers (12 Flashcards)

CH08-01 | Chapter 8 | Level 1 (Core Concept) | 📌 Final Exam Problem 1.1 & 1.2 (Page 32 - 4 Points)

Transformer Definition, Purpose & Structure

📌 **PROBLEM / QUESTION:**

What is an electrical transformer? What is its primary function and structure? Draw/describe its essential construction.



รูปประกอบ: Exam Problem 1.2: Transformer Core & Dual Windings

📐 **KEY FORMULAS & DEFINITIONS:**

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = a, \quad \frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{1}{a}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition & Function (ข้อ 1.1):
- A static electrical machine with NO moving parts that transfers electrical energy between circuits through electromagnetic induction.
- Purpose: Step-up (เพิ่มแรงดัน) or Step-down (ลดแรงดัน) AC voltage levels while keeping frequency constant.

2. Structure (ข้อ 1.2):
- Laminated ferromagnetic core (แกนเหล็กแผ่นซิลิคอนบางเรียงซ้อนกันเพื่อลดกระแสไหลวน Eddy Current)
- Primary winding (ขดลวดปฐมภูมิ N_1 ต่อเข้าแหล่งจ่าย)
- Secondary winding (ขดลวดทุติยภูมิ N_2 ต่อเข้าโหลด)

⚠️ **EXAM PITFALL & CAUTION:**

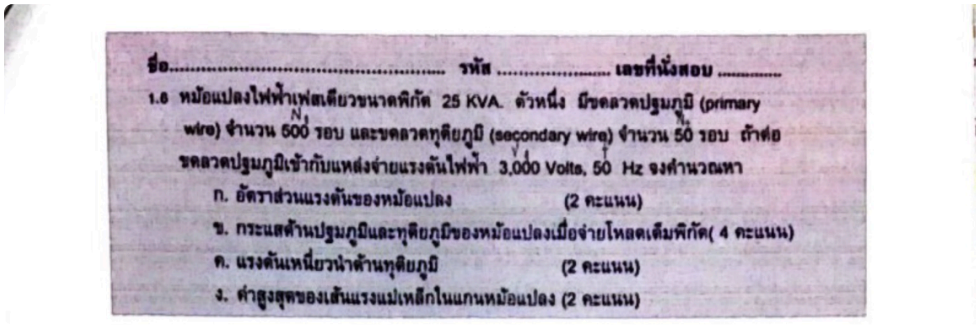
Transformers operate ONLY on Alternating Current (AC)! Never connect a transformer to a DC supply, as absence of changing flux ($d\phi/dt = 0$) induces zero back-EMF, causing excessive current that burns the primary winding.

CH08-02 | Chapter 8 | Level 2 (Exam / Calc) | Final Exam Problem 1.6 (Page 31 - 10 Points)

Single Phase 25 kVA Transformer Full Exam Problem

PROBLEM / QUESTION:

A single-phase transformer rated $25\ \text{kVA}$ has $N_1 = 500$ turns and $N_2 = 50$ turns. Primary winding is connected to $V_1 = 3,000\ \text{V}, 50\ \text{Hz}$. Calculate: (a) Turns ratio a, (b) Rated primary & secondary currents (I_1, I_2), (c) Induced secondary voltage V_2, (d) Maximum core flux Φ_m.


ชื่อ..... รหัส ..... เลขที่นั่งสอบ .....

1.6 หม้อแปลงไฟฟ้าเฟสเดียวขนาดพิกัด 25 KVA. ตัวหนึ่ง มีขดลวดปฐมภูมิ (primary wire) จำนวน 500 รอบ และขดลวดทุติยภูมิ (secondary wire) จำนวน 50 รอบ ถ้าต่อขดลวดปฐมภูมิเข้ากับแหล่งจ่ายแรงดันไฟฟ้า 3,000 Volts, 50 Hz จงคำนวณหา

ก. อัตราส่วนแรงดันของหม้อแปลง (2 คะแนน)

ข. กระแสด้านปฐมภูมิและทุติยภูมิของหม้อแปลงเมื่อจ่ายโหลดเต็มพิกัด(4 คะแนน)

ค. แรงดันเหนี่ยวนำด้านทุติยภูมิ (2 คะแนน)

ง. ค่าสูงสุดของเส้นแรงแม่เหล็กในแกนหม้อแปลง (2 คะแนน)

รูปประกอบ: Exam Problem 1.6: 25 kVA Transformer Full Calculation

KEY FORMULAS & DEFINITIONS:

$$a = \frac{N_1}{N_2}, \quad I_1 = \frac{S}{V_1}, \quad I_2 = \frac{S}{V_2}$$
$$V_2 = \frac{V_1}{a}, \quad V_1 = 4.44 f N_1 \Phi_m$$

STEP-BY-STEP MODEL SOLUTION:

1. (ก) Turns ratio (a):

$$a = \frac{N_1}{N_2} = \frac{500}{50} = 10$$

2. (ข) Rated currents:

$$I_1 = \frac{S}{V_1} = \frac{25,000}{3,000} = 8.33\ \text{A} = \frac{25}{3}\ \text{A}$$

$$I_2 = aI_1 = 10 \times 8.33 = 83.33\ \text{A} = \frac{250}{3}\ \text{A}$$

3. (ค) Secondary induced voltage (V_2):

$$V_2 = \frac{V_1}{a} = \frac{3,000}{10} = 300\ \text{V}$$

4. (ง) Maximum core flux (Φ_m):

$$V_1 = 4.44 f N_1 \Phi_m \Rightarrow 3,000 = 4.44(50)(500)\Phi_m$$

$$\Phi_m = \frac{3,000}{111,000} = 0.027\ \text{Wb} = 27.0\ \text{mWb}$$

EXAM PITFALL & CAUTION:

The constant 4.44 comes from $2\pi/\sqrt{2} \approx 4.44288$. Flux Φ_m is in Webers (Wb). Remember: V_1 uses N_1; V_2 uses N_2.

CH08-03 | Chapter 8 | Level 2 (Exam / Calc) | HW Ch.8 Problem 2 (Page 1)

Turns Ratio & Winding Calculations

PROBLEM / QUESTION:

The primary and secondary voltages of a single-phase transformer are $V_1 = 140\ \text{V}$ and $V_2 = 80\ \text{V}$. Primary current is $I_1 = 8\ \text{A}$. Secondary winding has $N_2 = 800$ turns. Find: (1) Primary winding turns N_1, (2) Secondary current I_2, (3) Turns ratio a.

KEY FORMULAS & DEFINITIONS:

$$a = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Turns Ratio (a):

$$a = \frac{V_1}{V_2} = \frac{140}{80} = 1.75$$

2. Primary turns (N_1):

$$N_1 = aN_2 = 1.75 \times 800 = 1,400\ \text{turns}$$

3. Secondary current (I_2):

$$I_2 = aI_1 = 1.75 \times 8\ \text{A} = 14.0\ \text{A}$$

EXAM PITFALL & CAUTION:

Voltage and turns are directly proportional ($V_1/V_2 = N_1/N_2$), but current is INVERSELY proportional ($I_1/I_2 = N_2/N_1$).

CH08-04 | Chapter 8 | Level 2 (Exam / Calc) | HW Ch.8 Problem 3 (Page 1)

200 kVA Transformer Voltage, Current & Core Flux

PROBLEM / QUESTION:

A single-phase transformer rated $200\ \text{kVA}, 2200\ \text{V}/220\ \text{V}, 50\ \text{Hz}$ has primary winding $N_1 = 200$ turns. Find: (1) Rated primary and secondary currents (I_1, I_2), (2) Secondary turns N_2, (3) Maximum magnetic core flux Φ_m.

KEY FORMULAS & DEFINITIONS:

$$I_1 = \frac{S}{V_1}, \quad I_2 = \frac{S}{V_2}$$
$$N_2 = \frac{V_2}{V_1} N_1, \quad \Phi_m = \frac{V_1}{4.44 f N_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Rated Currents:

$$I_1 = \frac{200 \times 10^3}{2,200} = 90.91\ \text{A}$$

$$I_2 = \frac{200 \times 10^3}{220} = 909.09\ \text{A}$$

2. Secondary turns (N_2):

$$a = \frac{2,200}{220} = 10 \Rightarrow N_2 = \frac{N_1}{10} = \frac{200}{10} = 20\ \text{turns}$$

3. Maximum flux (Φ_m):

$$\Phi_m = \frac{2,200}{4.44 \times 50 \times 200} = \frac{2,200}{44,400} = 0.0495\ \text{Wb} = 49.5\ \text{mWb}$$

EXAM PITFALL & CAUTION:

Always double check: $V_2 = 4.44 f N_2 \Phi_m$ must yield the EXACT same flux value: $220/(4.44 \times 50 \times 20) = 49.5\ \text{mWb}$.

CH08-05 | Chapter 8 | Level 2 (Exam / Calc) | HW Ch.8 Problem 4.4 (Page 1)

Transformer DC Connection Danger (Why Transformers Cannot Run on DC)

PROBLEM / QUESTION:

A $25\ \text{kVA}, 500/40\ \text{turns}$ transformer is supplied at primary by $300\ \text{V}, 50\ \text{Hz}$. (1) If we change to connect a DC voltage of $200\ \text{V}$ to the primary winding, what will the secondary current I_2 be? (2) What happens physically to the transformer?

KEY FORMULAS & DEFINITIONS:

$$e_2 = -N_2 \frac{d\Phi}{dt} = 0 \quad (\text{for steady DC})$$

STEP-BY-STEP MODEL SOLUTION:

1. Secondary current (I_2):

$$I_2 = 0\ \text{A}$$

Because DC voltage produces constant, steady magnetic flux ($\frac{d\Phi}{dt} = 0$), no voltage is induced in the secondary winding ($e_2 = 0\ \text{V}$), so $I_2 = 0\ \text{A}$.

2. Physical Consequence to Transformer:
With no alternating flux, there is zero back-EMF ($e_1 = 0$). Primary current is limited only by tiny copper resistance R_1 ($< 0.1\ \Omega$):

$$I_{1,\text{DC}} = \frac{200\ \text{V}}{R_1} \gg 1,000\ \text{A}!$$

The primary winding will overheat instantly, burn its enamel insulation, and catch fire!

EXAM PITFALL & CAUTION:

Classic trick question! Never write $I_2 = 200/a$. For DC, induced secondary voltage and current are ZERO, while primary burns.

CH08-06 | Chapter 8 | Level 1 (Core Concept) | 📌 Final Exam Problem 1.4 (Page 32 - 2 Points)

Ideal Transformer 4 Fundamental Hypotheses

📌 **PROBLEM / QUESTION:**

State the 4 fundamental properties/assumptions that define an Ideal Transformer (หม้อแปลงไฟฟ้าในอุดมคติ).

📐 **KEY FORMULAS & DEFINITIONS:**

Ideal: $R_1 = R_2 = 0, \quad \mu_r = \infty, \quad P_{\text{loss}} = 0, \quad \Phi_{\text{leakage}} = 0$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Zero Winding Resistance: $R_1 = 0, R_2 = 0$ (No I^2R copper losses).
2. Infinite Core Permeability: $\mu_r \to \infty$, so zero reluctance and zero magnetizing current needed ($I_m = 0$).
3. Zero Core Losses: No hysteresis loss and no eddy current loss ($P_{\text{core}} = 0$).
4. Zero Leakage Flux: All magnetic flux is perfectly confined within the core and links both windings completely ($X_1 = X_2 = 0$).

⚠️ **EXAM PITFALL & CAUTION:**

Real transformers have losses (efficiency $95 - 99\%$). An ideal transformer has 100% efficiency, so $S_{\text{in}} = S_{\text{out}}$ and $P_{\text{in}} = P_{\text{out}}$.

CH08-07 | Chapter 8 | Level 1 (Core Concept) | 📌 Final Exam Problem 1.5 (Page 32 - 2 Points)

Transformer Losses: Core Loss vs Copper Loss

📌 **PROBLEM / QUESTION:**

What types of power losses occur in a real transformer? Contrast Core Loss (Iron Loss) with Copper Loss (Winding Loss) regarding their cause and load dependency.

📐 **KEY FORMULAS & DEFINITIONS:**

$$P_{\text{total loss}} = P_{\text{core}} + P_{\text{cu}}$$
$$P_{\text{core}} = P_h + P_e \quad (\text{constant } 24/7)$$
$$P_{\text{cu}} = I_1^2 R_1 + I_2^2 R_2 \propto I_L^2$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Core Loss (Iron Loss / การสูญเสียในแกนเหล็ก):
- Hysteresis loss ($P_h \propto f B_{\text{max}}^{1.6}$) + Eddy current loss ($P_e \propto f^2 B_{\text{max}}^2 t^2$).
- Constant 24/7: Depends only on applied voltage and frequency; independent of load current!

2. Copper Loss (I^2R Loss / การสูญเสียในขดลวดทองแดง):
- Caused by Joule heating in primary and secondary winding resistances.
- Variable: Proportional to the square of load current ($P_{\text{cu}} \propto I_L^2$). Zero at no load!

⚠️ **EXAM PITFALL & CAUTION:**

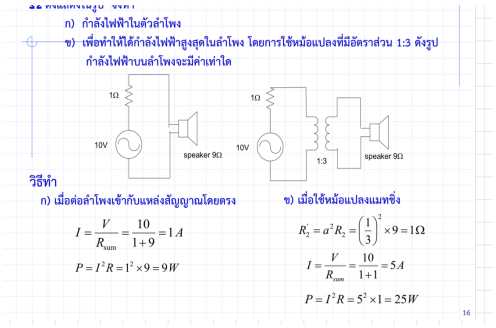
Maximum transformer efficiency occurs when Variable Loss equals Constant Loss: $P_{\text{cu}} = P_{\text{core}}$!

CH08-08 | Chapter 8 | Level 2 (Exam / Calc) | Ch07Transformer.pdf Example 1 (Page 18) & FunE P.7

Audio Transformer Impedance Matching for Maximum Power Transfer

PROBLEM / QUESTION:

A $10\ \text{V}$ audio source with internal source resistance $R_s = 1\ \Omega$ is connected to a speaker load $R_L = 9\ \Omega$. (a) Find power to the speaker without transformer. (b) If an ideal matching transformer with turns ratio $a = 1/3$ is inserted, find the new power transferred to the speaker.



รูปประกอบ: Audio Transformer Maximum Power Transfer Impedance Matching

KEY FORMULAS & DEFINITIONS:

$$R_L' = a^2 R_L = \left(\frac{N_1}{N_2}\right)^2 R_L$$

Maximum Power Transfer when $R_L' = R_s$

STEP-BY-STEP MODEL SOLUTION:

1. Without Transformer:

$$I = \frac{10}{1+9} = 1.0\ \text{A}$$

$$P_{\text{speaker}} = I^2 R_L = (1.0)^2 \times 9 = 9.0\ \text{W}$$

2. With Transformer ($a = 1/3$):
Reflected load resistance seen by source:

$$R_L' = a^2 R_L = \left(\frac{1}{3}\right)^2 \times 9 = \frac{1}{9} \times 9 = 1.0\ \Omega$$

Notice $R_L' = R_s = 1.0\ \Omega$ (perfect Maximum Power Transfer match!).
New primary current:

$$I_1 = \frac{10}{1+1} = 5.0\ \text{A}$$

Power delivered to reflected load (and thus speaker):

$$P_{\text{speaker}} = I_1^2 R_L' = (5.0)^2 \times 1.0 = 25.0\ \text{W}$$

Power increases dramatically from $9\ \text{W}$ to $25\ \text{W}$!

EXAM PITFALL & CAUTION:

Reflected impedance scales as a^2: $Z_1 = a^2 Z_2$. In audio systems, transformers match unequal impedances to maximize transferred power.

CH08-09 | Chapter 8 | Level 1 (Core Concept) | Lecture Slides & Formula Sheet

Open-Circuit (No-Load) Test for Shunt Branch Parameters

PROBLEM / QUESTION:

In a transformer Open-Circuit Test: (1) Which winding is energized and which is left open? (2) Which equivalent circuit parameters are determined, and what does the wattmeter reading represent?

KEY FORMULAS & DEFINITIONS:

$$P_{oc} \approx P_{\text{core}}, \quad R_c = \frac{V_{oc}^2}{P_{oc}}$$

$$I_c = \frac{V_{oc}}{R_c}, \quad I_m = \sqrt{I_{oc}^2 - I_c^2}, \quad X_m = \frac{V_{oc}}{I_m}$$

STEP-BY-STEP MODEL SOLUTION:

1. Connection:
- Low Voltage (LV) side is energized at rated voltage.
- High Voltage (HV) side is left completely OPEN for operator safety.

2. Measured Values & Parameters:
- Wattmeter measures Core Loss directly ($P_{oc} = P_{\text{core}}$) because no-load current is tiny ($< 3-5\%$), making winding I^2R negligible.
- Yields shunt excitation branch parameters: Core resistance R_c (modeling hysteresis & eddy losses) and magnetizing reactance X_m (modeling core magnetization).

EXAM PITFALL & CAUTION:

Open-Circuit test yields CORE LOSS (P_{core}) and shunt branch (R_c, X_m). Do NOT confuse with short-circuit test!

CH08-10 | Chapter 8 | Level 1 (Core Concept) | Lecture Slides & Formula Sheet

Short-Circuit Test for Series Winding Parameters

PROBLEM / QUESTION:

In a transformer Short-Circuit Test: (1) Which winding is short-circuited and which is energized? (2) Which equivalent circuit parameters are determined, and what does the wattmeter reading represent?

KEY FORMULAS & DEFINITIONS:

$$P_{sc} \approx P_{\text{cu}}, \quad R_{eq} = \frac{P_{sc}}{I_{sc}^2}$$

$$Z_{eq} = \frac{V_{sc}}{I_{sc}}, \quad X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

STEP-BY-STEP MODEL SOLUTION:

1. Connection:
- Low Voltage (LV) side is securely SHORT-CIRCUITED with a thick jumper bar.
- High Voltage (HV) side is energized with a REDUCED voltage ($5-10\%$ of rated) adjusted until rated current flows.

2. Measured Values & Parameters:
- Wattmeter measures Full-Load Copper Loss ($P_{sc} = P_{\text{cu}}$) because applied voltage is so low that core flux and core loss are negligible.
- Yields series equivalent parameters: Equivalent resistance $R_{eq} = R_1 + a^2 R_2$ and equivalent leakage reactance $X_{eq} = X_1 + a^2 X_2$.

EXAM PITFALL & CAUTION:

Short-Circuit test yields COPPER LOSS (P_{cu}) and series parameters (R_{eq}, X_{eq}). Never apply full rated voltage during a short-circuit test!

CH08-11 | Chapter 8 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Transformer Voltage Regulation Formula & Power Factor Dependency

📌 **PROBLEM / QUESTION:**

State the formula for Transformer Voltage Regulation ($VR\%$). How does the power factor (lagging vs leading) affect voltage regulation, and can $VR\%$ ever be negative?

📐 **KEY FORMULAS & DEFINITIONS:**

$$VR\% = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

Approx:

$$VR\% \approx \frac{I_2 R_{eq2} \cos\theta \pm I_2 X_{eq2} \sin\theta}{V_2} \times 100\%$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Formula:

$$VR\% = \frac{V_{2,NL} - V_{2,FL}}{V_{2,FL}} \times 100\%$$

2. Effect of Power Factor:

- Lagging PF (Inductive load): (+) sign applies. Voltage drops under load ($V_{FL} < V_{NL}$), so $VR\% > 0$ (positive).
- Unity PF ($PF = 1.0$): Small positive $VR\%$.
- Leading PF (Capacitive load): (-) sign applies. Due to capacitive Ferranti effect, terminal voltage rises under load ($V_{FL} > V_{NL}$), causing $VR\% < 0$ (NEGATIVE)!

⚠️ **EXAM PITFALL & CAUTION:**

Negative voltage regulation ($VR\% < 0$) happens ONLY with LEADING power factors (capacitive loads), meaning voltage at full load is HIGHER than at no load!

CH08-12 | Chapter 8 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Auto-Transformer Principle, Higher kVA Rating & Common Winding Risk

📌 **PROBLEM / QUESTION:**

What is an Auto-Transformer? Why does it possess a higher kVA power rating and higher efficiency than a conventional two-winding transformer of the same physical size, and what is its primary safety drawback?

📐 **KEY FORMULAS & DEFINITIONS:**

$$\frac{S_{\text{auto}}}{S_{\text{2-w}}} = \frac{1}{1 - 1/a_{\text{auto}}}$$

Conducted power + Transformed power

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Construction & Principle:
An auto-transformer has a SINGLE continuous winding shared partly by both primary and secondary circuits (tapped common winding).

2. Higher Rating & Efficiency:
Power is transferred via two mechanisms simultaneously:
- By electromagnetic induction (transformed power)
- By direct electrical conduction (conducted power)
Because part of the energy flows conductively through the shared copper, magnetic core and windings can be much smaller and lighter, with lower I^2R and core losses.

3. Safety Drawback:
NO ELECTRICAL ISOLATION between primary (HV) and secondary (LV)! If the common ground connection opens, the full high voltage appears directly on the low voltage terminals, risking electrocution.

⚠️ **EXAM PITFALL & CAUTION:**

Never use an autotransformer where human safety isolation from high voltage mains is required (e.g. medical equipment or bathroom supplies).

PART 4

Chapter 9: Power Transmission Systems & Energy Conservation

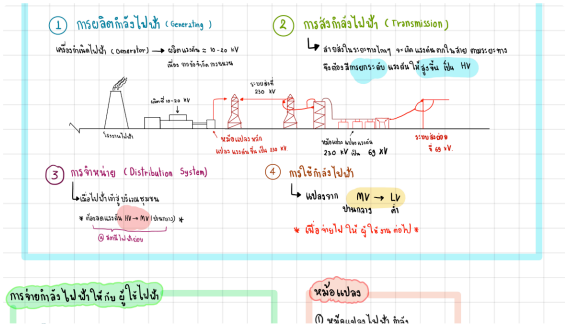
Thai Grid Voltage Stages, EGAT/MEA/PEA, Power Plants, Tower Classifications, Energy Conservation (10 Flashcards)

CH09-01 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.3 (หน้า 25 - 4 คะแนน)

Structure of Electrical Power System & Voltage Levels

📌 PROBLEM / QUESTION:

State the 4 main stages of an electric power system, and classify the following standard grid voltages in Thailand: 500 kV, 22 kV, 380 V, and residential household voltage.



รูปประกอบ: Electric Power System Stages (Generation to Utilization)

📐 KEY FORMULAS & DEFINITIONS:

$$\text{Stages: Generation} \rightarrow \text{Transmission} \rightarrow \text{Distribution} \rightarrow \text{Utilization}$$

$$P = \sqrt{3} V_L I_L \cos\theta \implies I_L = \frac{P}{\sqrt{3} V_L \cos\theta}$$

$$P_{\text{loss}} = 3 I_L^2 R_{\text{line}} \propto \frac{1}{V_L^2}$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Four Stages of Power System:
- Generation: 10 - 20 kV (alternator output terminals)
- Transmission: 115 kV, 230 kV, 500 kV (long-distance bulk power transfer)
- Distribution: 22 kV, 24 kV, 33 kV (medium voltage to substations and streets)
- Utilization: 380 V (3-phase) / 220 V (1-phase) (end consumers)

2. Voltage Classifications for Exam:
- 500 kV: Extra High Voltage (EHV)
- 22 kV: Medium Voltage (MV) - standard PEA/MEA primary distribution
- 380 V: Low Voltage (LV) - standard 3-phase line-to-line utilization
- Household: 220 V (or modern 230 V) - single-phase line-to-neutral utilization.

⚠️ EXAM PITFALL & CAUTION:

Why transmit at Extra High Voltage (500 kV)? Because stepping up VL reduces line current IL proportionally, which decreases transmission line I^2 R loss by a factor of (1/V)^2.

CH09-02 | Chapter 9 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam Cheatsheet

Three Electricity Authorities in Thailand & Roles

📌 **PROBLEM / QUESTION:**

Name the 3 state electricity authorities in Thailand and explain their specific responsibilities and geographical jurisdictions.

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{EGAT: Generation} + \text{High-Voltage Grid (Nationwide)}$$

$$\text{MEA: Distribution in 3 Provinces (BKK, Nonthaburi, Samut Prakan)}$$

$$\text{PEA: Distribution in 74 Provinces (All other provinces)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. EGAT (Electricity Generating Authority of Thailand / กฟผ.):
Responsible for bulk power generation (hydro, thermal, combined cycle) and the nationwide high-voltage transmission grid (115 kV - 500 kV). Sells wholesale electricity to MEA, PEA, and large direct industrial customers.

2. MEA (Metropolitan Electricity Authority / กฟน.):
Responsible for electricity distribution and retail customer service strictly across 3 provinces: Bangkok, Nonthaburi, and Samut Prakan.

3. PEA (Provincial Electricity Authority / กฟภ.):
Responsible for electricity distribution and retail customer service across all remaining 74 provinces of Thailand.

⚠️ **EXAM PITFALL & CAUTION:**

Exam trap: MEA covers strictly 3 provinces (Bangkok, Nonthaburi, Samut Prakan). All other 74 provinces belong to PEA.

CH09-03 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.1 (หน้า 25 - 2 คะแนน)

Fuel Mix for Power Generation in Thailand

📌 **PROBLEM / QUESTION:**

What is the primary fuel source used most heavily for electricity generation in Thailand, and what fuel source is used the least?

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Primary / Most Used: Natural Gas (ก๊าซธรรมชาติ)} \approx 55\% - 60\%$$

$$\text{Second: Coal / Lignite (ถ่านหินลิกไนต์)} + \text{Imported Hydro (สปป.ลาว)}$$

$$\text{Least Used: Heavy Fuel Oil / Diesel (น้ำมันเตา/ดีเซล)} < 1\%$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Most Used Fuel:
Natural Gas (ก๊าซธรรมชาติ) generates over 55-60% of total electrical energy in Thailand, supplied from the Gulf of Thailand, Myanmar pipelines, and imported Liquefied Natural Gas (LNG).

2. Secondary Sources:
Coal / Lignite (ถ่านหินลิกไนต์ แม่เมาะ) and imported hydropower from Lao PDR.

3. Least Used Fuel:
Heavy fuel oil (น้ำมันเตา) and diesel, utilized only as emergency backup during gas pipeline maintenance due to exceptionally high operating costs and particulate emissions.

⚠️ **EXAM PITFALL & CAUTION:**

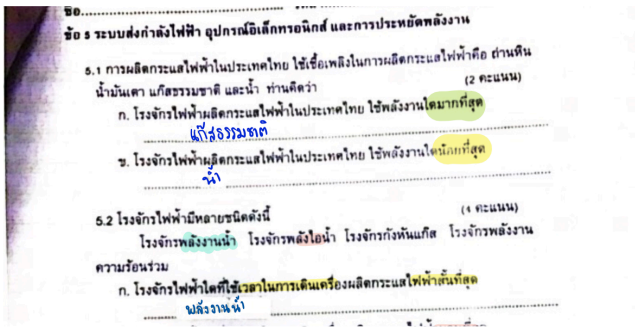
Do NOT answer solar, hydro, or petroleum oil as Thailand's primary fuel. It is Natural Gas by a large margin.

CH09-04 | Chapter 9 | Level 2 (Exam / Calc) | 📌 Final Exam ข้อ 5.2 (หน้า 25 - 4 คะแนน)

Base Load vs Peak Load Power Plants & Startup Times

📌 **PROBLEM / QUESTION:**

Compare Base Load and Peak Load power plants regarding operating objectives, fuel types, capital vs operating costs, and startup times. Give standard plant examples.


ข้อ 5 ระบบส่งกำลังไฟฟ้า อุปกรณ์อิเล็กทรอนิกส์ และการประหยัดพลังงาน

5.1 การผลิตกระแสไฟฟ้าในประเทศไทย ใช้เชื้อเพลิงในการผลิตกระแสไฟฟ้าคือ ถ่านหิน น้ำมันเตา แก๊สธรรมชาติ และน้ำ ท่านคิดว่า (2 คะแนน)

ก. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดมากที่สุด แก๊สธรรมชาติ

ข. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดน้อยที่สุด น้ำ

5.2 โรงจักรไฟฟ้ามีหลายชนิดดังนี้ (4 คะแนน)

โรงจักรพลังงานน้ำ โรงจักรพลังไอน้ำ โรงจักรกังหันแก๊ส โรงจักรพลังงานความร้อนร่วม

ก. โรงจักรไฟฟ้าใดที่ใช้เวลาในการเดินเครื่องผลิตกระแสไฟฟ้าสั้นที่สุด พลังงานน้ำ

รูปประกอบ: Exam Problem 5.1 & 5.2: Power Plants in Thailand

📐 **KEY FORMULAS & DEFINITIONS:**

Base Load: Continuous 24/7 operation, High Capital Cost, Low Fuel/Operating Cost

Peak Load: Fast Startup (minutes), Low Capital Cost, High Fuel/Operating Cost

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Base Load Power Plants:
- Purpose: Run continuously 24 hours a day at rated constant capacity to meet baseline national demand.
- Plant Types: Coal/Lignite Thermal Plants (Mae Moh), Nuclear, Combined-Cycle Gas Turbine (CCGT).
- Characteristics: Very high capital construction cost, low fuel/operating cost per kWh, very slow startup time (hours to days from cold start).

2. Peak Load Power Plants:
- Purpose: Operated temporarily during peak demand hours (typically hot weekday afternoons and early evenings) and sudden grid disturbances.
- Plant Types: Gas Turbines (Open Cycle), Pumped-Storage Hydro (เขื่อนลำตะคองชลภาวัฒนา), Conventional Hydro.
- Characteristics: Lower capital investment, high fuel operating cost per kWh, extremely fast startup time (starts and synchronizes to grid in 3 - 15 minutes).

⚠️ **EXAM PITFALL & CAUTION:**

Hydro with large reservoir or pumped storage is ideal for peak load because water valves can be opened to deliver full megawatts in under 5 minutes.

CH09-05 | Chapter 9 | Level 1 (Core Concept) | Lecture Chapter 9 (Power Systems)

Transmission Tower Classifications & Structures

PROBLEM / QUESTION:

Classify transmission tower structures into Suspension, Tension, Dead-end, and Transposition towers, stating the mechanical and electrical purpose of each.



รูปประกอบ: High-Voltage Transmission Towers (Suspension, Tension, Transposition)

KEY FORMULAS & DEFINITIONS:

Suspension: Straight paths, supports vertical conductor weight

Tension / Angle: Turns/corners, withstands directional pull tension

Dead-end: Terminal stations, handles one-sided full mechanical tension

Transposition: Balances mutual inductance $L_{12} = L_{23} = L_{31}$ and capacitance across 3 phases

STEP-BY-STEP MODEL SOLUTION:

1. Suspension Towers (เสาแขวน):
Used along straight line routes. Insulators hang vertically downwards to support only the physical downward weight of conductors without taking longitudinal line tension.

2. Tension / Angle Towers (เสารั้ง / เสามุม):
Installed where transmission line changes direction (curves/corners). Insulators are placed horizontally in tension to withstand the horizontal mechanical pulling vector.

3. Dead-end / Terminal Towers (เสาสิ้นสุด):
Placed where the line terminates at a substation or crosses major obstacles (wide rivers, mountains). Engineered to withstand the full, unbalanced tension from one side.

4. Transposition Towers (เสาสลับเฟส):
Swaps physical conductor positions at regular 1/3 distance intervals to equalize line inductances and capacitances, balancing phase voltages.

EXAM PITFALL & CAUTION:

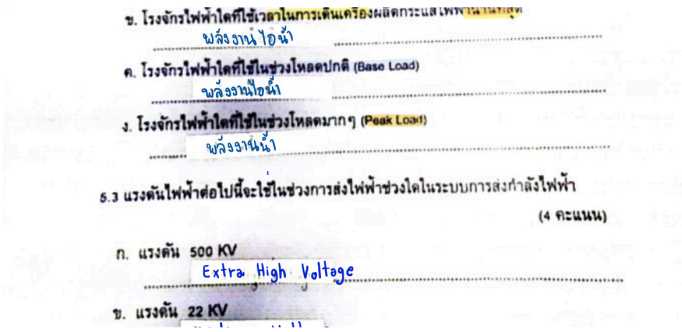
Transposition is done solely for electrical symmetry (equalizing mutual inductance and capacitance), preventing unbalance on long 3-phase lines.

CH09-06 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.4 (หน้า 25 - 4 คะแนน)

Customer Categories & Supply Voltage Thresholds

📌 **PROBLEM / QUESTION:**

State the standard customer categories used by MEA and PEA (Residential, Small General Service, Medium/Large General Service), and explain why large industrial customers must receive high voltage (22 kV - 115 kV) rather than low voltage (380 V).



รูปประกอบ: Exam Problem 5.3 & 5.4: Grid Voltage Classifications

📐 **KEY FORMULAS & DEFINITIONS:**

$$I = \frac{P}{\sqrt{3} V_L \cos\theta}$$

$$\text{If } P = 10 \text{ MW at } 380 \text{ V: } I \approx 18,200 \text{ A (Impossibly thick cables)}$$

$$\text{If } P = 10 \text{ MW at } 22 \text{ kV: } I \approx 315 \text{ A (Standard manageable cable)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Customer Categories (MEA/PEA):
- Type 1: Residential (บ้านอยู่อาศัย) - Single-phase 220 V or 3-phase 380 V for large homes.
- Type 2: Small General Service (ธุรกิจขนาดเล็ก) - Demand < 30 kW (shops, small offices).
- Type 3: Medium General Service (ธุรกิจขนาดกลาง) - Demand 30 kW - 999 kW.
- Type 4: Large General Service (ธุรกิจขนาดใหญ่) - Demand >= 1,000 kW (factories, malls).

2. Why Large Customers Take High Voltage (22 kV - 115 kV):
- Current Reduction: From P = sqrt(3)*V*I*cos(theta), supplying massive power at 380 V would require tens of thousands of amperes, needing impossibly thick copper busbars and incurring massive voltage drops (Delta V = I*Z).
- Grid Reliability: Directly feeding from 22 kV or 115 kV keeps the customer's heavy fluctuating loads from distorting voltage quality for neighboring residential consumers on the low-voltage network.

⚠️ **EXAM PITFALL & CAUTION:**

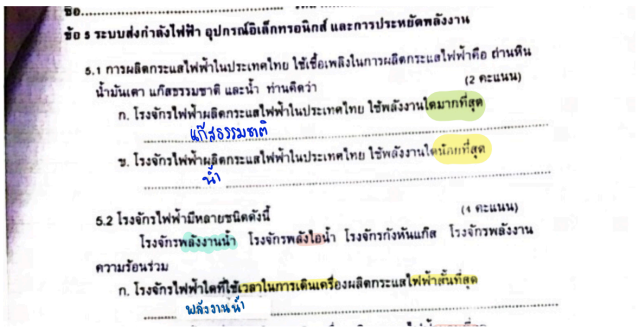
Supplying large loads at low voltage causes extreme copper loss I^2 R and severe voltage drop Delta V = I*Z.

CH09-07 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.2 (หน้า 25 - 2 คะแนน) & Sheet ติว

Power Plant Specifics in Thailand (Bhumibol, Ubolratana, Mae Moh, Nam Pong)

📌 **PROBLEM / QUESTION:**

Identify the power generation types and fuels of these major Thai power plants: Bhumibol Dam, Ubolratana Dam, Mae Moh Power Plant, and Nam Pong Power Plant.


ข้อ 5 ระบบส่งกำลังไฟฟ้า อุปกรณ์อิเล็กทรอนิกส์ และการประหยัดพลังงาน

5.1 การผลิตกระแสไฟฟ้าในประเทศไทย ใช้เชื้อเพลิงในการผลิตกระแสไฟฟ้าคือ ถ่านหิน น้ำมันเตา แก๊สธรรมชาติ และน้ำ ท่านคิดว่า (2 คะแนน)

ก. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดมากที่สุด แก๊สธรรมชาติ

ข. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดน้อยที่สุด น้ำ

5.2 โรงจักรไฟฟ้ามีหลายชนิดดังนี้ (1 คะแนน)

โรงจักรพลังงานน้ำ โรงจักรพลังไอน้ำ โรงจักรกังหันแก๊ส โรงจักรพลังงานความร้อนร่วม

ก. โรงจักรไฟฟ้าใดที่ใช้เวลาในการเดินเครื่องผลิตกระแสไฟฟ้าสั้นที่สุด พลังงานน้ำ

รูปประกอบ: Exam Problem 5.1 & 5.2: Major Power Plants in Thailand

📐 **KEY FORMULAS & DEFINITIONS:**

Bhumibol / Ubolratana: Hydroelectric (พลังน้ำ)

Mae Moh (Lampang): Lignite Coal Thermal (พลังความร้อนถ่านหินลิกไนต์)

Nam Pong (Khon Kaen): Natural Gas Combined Cycle (พลังความร้อนร่วมก๊าซธรรมชาติ)

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Bhumibol Dam (เขื่อนภูมิพล - Tak):
Hydroelectric power plant (พลังน้ำ); Thailand's largest concrete arch-gravity dam; also features pumped-storage units.

2. Ubolratana Dam (เขื่อนอุบลรัตน์ - Khon Kaen):
Hydroelectric multi-purpose dam (พลังน้ำ) supplying renewable power and irrigation in the Northeast (Isan).

3. Mae Moh Power Plant (โรงไฟฟ้าแม่เมาะ - Lampang):
Lignite coal-fired thermal power plant (โรงไฟฟ้าพลังความร้อนลิกไนต์); largest coal power complex in SE Asia, operating as a critical base load plant.

4. Nam Pong Power Plant (โรงไฟฟ้าน้ำพอง - Khon Kaen):
Combined-Cycle gas turbine power plant (โรงไฟฟ้าพลังความร้อนร่วม) fueled primarily by domestic natural gas from Sinphuhorm gas field.

⚠️ **EXAM PITFALL & CAUTION:**

Do not confuse Mae Moh (Lignite Coal in Lampang) with Nam Pong (Natural Gas in Khon Kaen).

CH09-08 | Chapter 9 | Level 1 (Core Concept) | Lecture Chapter 9 (Renewable Energy)

Residential Solar PV Rooftop System Components

PROBLEM / QUESTION:

List the core components of a grid-tied residential Solar PV Rooftop system and explain the essential safety role of Anti-Islanding protection.

KEY FORMULAS & DEFINITIONS:

$$\text{PV Panels (DC)} \rightarrow \text{DC Combiner/Breaker} \rightarrow \text{Grid-Tied Inverter (DC to AC)} \rightarrow \text{Bi-directional Meter} \rightarrow \text{Grid}$$

$$\text{Anti-Islanding: Grid Loss} \implies \text{Inverter disconnects immediately within 2 seconds}$$

STEP-BY-STEP MODEL SOLUTION:

1. Core Components:
- Solar PV Modules (แผงโซลาร์เซลล์): Converts sunlight photons into DC electricity.
- DC Protection Box (ตู้ไฟ DC): Fuses, DC Circuit Breaker, DC Surge Protection Device (SPD).
- Grid-Tied Inverter (อินเวอร์เตอร์): Converts variable DC into synchronized 220 V 50 Hz AC matching the utility grid.
- AC Protection Box (ตู้ไฟ AC): AC Breaker, RCBO/RCD, AC SPD.
- Bi-directional Meter (มิเตอร์ 2 ทาง): Measures both imported energy from grid and surplus exported solar energy.

2. Anti-Islanding Protection (Mandatory):
When the public utility grid suffers a blackout or is switched off for line maintenance, the inverter MUST disconnect automatically within 2 seconds. This prevents backfeeding power into downed lines, protecting utility repair technicians from lethal electrocution.

EXAM PITFALL & CAUTION:

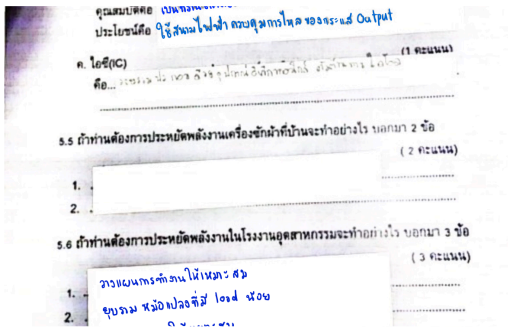
Anti-Islanding is a strict legal requirement by MEA/PEA for all grid-tied solar installations.

CH09-09 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.5 (หน้า 24 - 2 คะแนน)

Energy Conservation in Household Washing Machines

📌 **PROBLEM / QUESTION:**

State practical energy-saving guidelines when using a residential washing machine.



รูปประกอบ: Exam Problem 5.5: Household Energy Conservation

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Wash at full rated load} + \text{Use cold water} + \text{Select Inverter Motor}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Wash at Full Capacity:
Accumulate clothes and wash at the machine's rated full drum capacity, rather than doing frequent small washes, reducing total motor running cycles and water consumption.

2. Use Cold Water:
Use cold water setting (30°C or room temp) instead of hot water heating cycles (60°C - 90°C), as electric water heating elements consume over 80-90% of total wash cycle energy.

3. Pre-soak Heavy Stains:
Pre-soak heavily soiled clothes manually before putting them in, allowing standard or quick wash modes rather than heavy prolonged cycles.

4. Use Inverter Motor & Air Dry:
Choose washing machines with high energy-efficiency labels (ฉลากประหยัดไฟเบอร์ 5 ดาวสูง) with Inverter BLDC motors, and utilize natural outdoor sunlight for drying instead of electric tumble dryers.

⚠️ **EXAM PITFALL & CAUTION:**

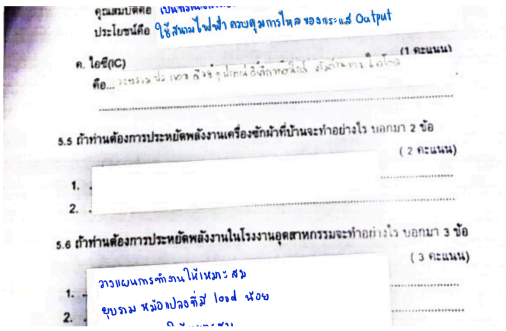
Electric water heating in washing machines consumes far more energy than the motor itself. Avoid hot wash cycles unless medically required.

CH09-10 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.6 (หน้า 24 - 2 คะแนน)

Energy Conservation in Industrial Manufacturing Plants

📌 **PROBLEM / QUESTION:**

Propose practical energy conservation measures for an industrial factory across: (a) Electric motors and drives, (b) Lighting systems, and (c) Air conditioning and thermal systems.



รูปประกอบ: Exam Problem 5.6: Industrial Factory Energy Conservation

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Motor: VFD / Inverter} + \text{IE3/IE4 Motors} \quad \Delta P_{\text{save}} \propto (1 - (n_2/n_1)^3)$$

$$\text{Power Factor Correction: Install Capacitor Banks} \implies PF \geq 0.85$$

$$\text{Lighting: Replace HID/Fluorescent with LED} \implies 50\% - 70\% \text{ reduction}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Electric Motor Systems (Consumes > 65% of industrial power):
- Install Variable Frequency Drives (VFD) on centrifugal pumps and fans to adjust motor speed according to real-time load instead of wasteful mechanical throttling valves (affinity laws: P proportional to n^3).
- Upgrade old motors to Premium High-Efficiency Motors (IE3 or IE4 standard).
- Correct Plant Power Factor by installing Automatic Capacitor Banks at the main switchboard (MDB) to maintain PF >= 0.85, eliminating PEA/MEA reactive power fines (ค่า kVAR).

2. Lighting Systems:
- Retrofit older fluorescent (T8) and high-intensity discharge (HID/Metal Halide) lamps with industrial LED luminaires (saving 50-70% power).
- Install smart daylight sensors and occupancy sensors in warehouses.

3. Air Conditioning & Chillers:
- Optimize chilled water temperatures and schedule chiller sequencing based on production shifts.
- Maintain and clean chiller condenser tubes and cooling tower fins regularly to prevent heat-transfer scale buildup.

⚠️ **EXAM PITFALL & CAUTION:**

Capacitor banks do NOT reduce active power (kW) consumed by the load, but they eliminate reactive power (kVAR) from the grid, reducing line current and avoiding utility penalty fees.

PART 5

Chapter 10: Measuring Instruments & Sensors

PMMC Principles, DC Ammeters & Voltmeters with CDRX, Ayrton Shunts, Ohmmeters, Wattmeters, Accuracy (14 Flashcards)

CH10-01 | Chapter 10 | Level 1 (Core Concept) | 📌 Homework Chapter 10 & Final Exam ข้อ 3

PMMC / Galvanometer Working Principle & Deflection Torque

📌 **PROBLEM / QUESTION:**

Explain the operating principle of a Permanent Magnet Moving Coil (PMMC / D'Arsonval) instrument. State the formula for deflecting torque and explain why PMMC instruments only measure DC directly.

📐 **KEY FORMULAS & DEFINITIONS:**

$$T_d = B \cdot I \cdot N \cdot A = G \cdot I$$

$$T_c = C \cdot \theta$$

$$\text{At equilibrium: } T_d = T_c \implies \theta = \left(\frac{G}{C}\right) I \propto I$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Operating Principle:
A rectangular copper coil carrying current I is pivoted inside the magnetic field B of a permanent magnet. The interaction of current and magnetic flux creates a Lorentz force (F = I*L x B), producing a deflecting torque Td.

2. Linear Scale:
Since deflecting torque Td = B*I*N*A and restoring spring torque Tc = C*theta, at steady state:
theta = (B*N*A / C) * I
Because B, N, A, and C are constant, the pointer deflection angle theta is directly proportional to current I, yielding a perfectly LINEAR scale.

3. Why DC Only:
If connected to alternating current (AC), the torque alternates direction every half-cycle (50 Hz). Due to the mechanical inertia of the moving coil and pointer, the meter cannot follow this rapid oscillation and indicates strictly the AVERAGE value, which is ZERO for pure sinusoidal AC.

⚠️ **EXAM PITFALL & CAUTION:**

PMMC measures strictly average DC. To measure AC with a PMMC meter, a rectifier circuit (diodes) must be added beforehand.

CH10-02 | Chapter 10 | Level 2 (Exam / Calc) | 📌 Homework Chapter 10 ข้อ 1 & Past Exam

PMMC Torque & Deflection Calculation

📌 **PROBLEM / QUESTION:**

A PMMC coil has N = 100 turns, width w = 2 cm, length l = 3 cm, and sits in a uniform radial flux density B = 0.2 T. The spring constant is C = 1.2 × 10^-6 N*m/degree. Calculate the deflecting torque and pointer deflection angle when carrying a current of 1 mA.

📐 **KEY FORMULAS & DEFINITIONS:**

$$A = w \cdot l = (0.02)(0.03) = 6 \times 10^{-4}\ \text{m}^2$$

$$T_d = B \cdot I \cdot N \cdot A$$

$$\theta = \frac{T_d}{C}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Calculate Coil Area:
A = 0.02 m * 0.03 m = 6 × 10^-4 m^2

2. Calculate Deflecting Torque (Td):
Td = B * I * N * A
Td = (0.2 T) * (1 × 10^-3 A) * (100 turns) * (6 × 10^-4 m^2)
Td = 1.2 × 10^-5 N*m

3. Calculate Deflection Angle (theta):
At equilibrium, Td = Tc = C * theta
theta = Td / C
theta = (1.2 × 10^-5 N*m) / (1.2 × 10^-6 N*m/degree)
theta = 10 degrees.

⚠️ **EXAM PITFALL & CAUTION:**

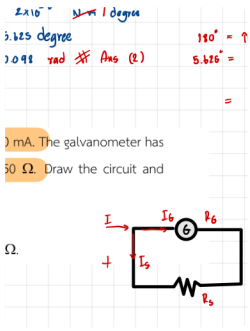
Always convert dimensions from cm to meters (m) to avoid calculation errors by factors of 10,000.

CH10-03 | Chapter 10 | Level 2 (Exam / Calc) | Homework Chapter 10 ข้อ 2 & Final Exam

DC Ammeter Design with and without CDRX

PROBLEM / QUESTION:

A Galvanometer has internal resistance Rm = 100 ohms, full-scale current Ifsd = 1 mA, and Critical Damping Resistance CDRX = 900 ohms. Design a DC Ammeter to measure full scale I = 1 A for:
(a) Pure circuit without CDRX
(b) Circuit with CDRX (damping resistor).



รูปประกอบ: Homework 10 Problem 2: DC Ammeter Design with and without CDRX

KEY FORMULAS & DEFINITIONS:

$$\text{Without CDRX: } R_{\text{sh}} = \frac{I_{\text{fsd}} R_m}{I - I_{\text{fsd}}} = \frac{R_m}{m - 1}$$

$$\text{With CDRX: Series resistor } R_s = \text{CDRX} - R_m$$

$$R'_{\text{sh}} = \frac{I_{\text{fsd}}(R_m + R_s)}{I - I_{\text{fsd}}} = \frac{I_{\text{fsd}} \cdot \text{CDRX}}{I - I_{\text{fsd}}}$$

$$\text{Total Meter Resistance: } R_{\text{in}} = R_{\text{sh}} \parallel (R_m + R_s)$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Without CDRX:
- Shunt resistor Rsh is connected in parallel with Rm:
I_sh = I - Ifsd = 1 A - 0.001 A = 0.999 A
Rsh = (Ifsd * Rm) / (I - Ifsd) = (0.001 * 100) / 0.999 = 0.1001 ohms.

2. Part (b) With CDRX:
- To achieve critical damping, the total resistance across meter terminals must equal CDRX = 900 ohms.
- Add series resistor Rs = CDRX - Rm = 900 - 100 = 800 ohms.
- Now the effective meter branch resistance is (Rm + Rs) = CDRX = 900 ohms.
- New Shunt Resistor Rsh':
Rsh' = (Ifsd * CDRX) / (I - Ifsd) = (0.001 * 900) / 0.999 = 0.9009 ohms.

EXAM PITFALL & CAUTION:

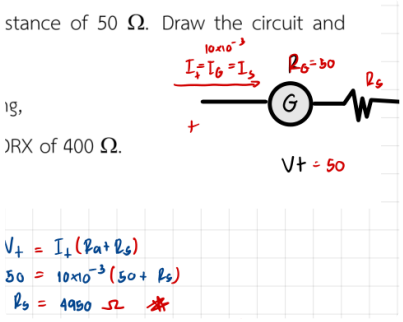
Do not connect CDRX in parallel with Rm! Rs is connected in SERIES with Rm so their sum equals CDRX, and then Rsh is connected in parallel with the entire combination.

CH10-04 | Chapter 10 | Level 2 (Exam / Calc) | Homework Chapter 10 ข้อ 3 & Final Exam

DC Voltmeter Design with and without CDRX

PROBLEM / QUESTION:

A Galvanometer has Rm = 100 ohms, Ifsd = 1 mA, and CDRX = 900 ohms. Design a DC Voltmeter to measure full-scale V = 100 V for:
(a) Pure circuit without CDRX
(b) Circuit with CDRX (damping resistor).



รูปประกอบ: Homework 10 Problem 3: DC Voltmeter Design with and without CDRX

KEY FORMULAS & DEFINITIONS:

$$\text{Without CDRX: } R_s = \frac{V}{I_{\text{fsd}}} - R_m$$

$$\text{With CDRX: Shunt damping resistor } R_p = \text{CDRX} = 900\ \Omega$$

$$I_{\text{total}} = I_{\text{fsd}} + I_p = I_{\text{fsd}} \left(1 + \frac{R_m}{R_p}\right)$$

$$R_s' = \frac{V - V_m}{I_{\text{total}}} = \frac{V - I_{\text{fsd}} R_m}{I_{\text{total}}}$$

$$\text{Sensitivity: } S = \frac{1}{I_{\text{fsd}}}\ (\Omega/\text{V})$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Without CDRX:
- Total required resistance R_total = V / Ifsd = 100 V / 0.001 A = 100,000 ohms (100 k-ohms).
- Multiplier series resistor Rs = R_total - Rm = 100,000 - 100 = 99,900 ohms (99.9 k-ohms).

2. Part (b) With CDRX:
- To provide critical damping in a voltmeter, a parallel damping resistor Rp = CDRX = 900 ohms is connected in PARALLEL with Rm.
- Voltage drop across coil at full scale: Vm = Ifsd * Rm = 0.001 A * 100 ohms = 0.1 V.
- Current through Rp: Ip = Vm / Rp = 0.1 V / 900 ohms = 0.1111 mA.
- Total circuit current: I_total = Ifsd + Ip = 1.0 mA + 0.1111 mA = 1.1111 mA.
- Multiplier series resistor Rs':
Rs' = (V - Vm) / I_total = (100 V - 0.1 V) / (1.1111 × 10^-3 A) = 89,910 ohms (89.91 k-ohms).

EXAM PITFALL & CAUTION:

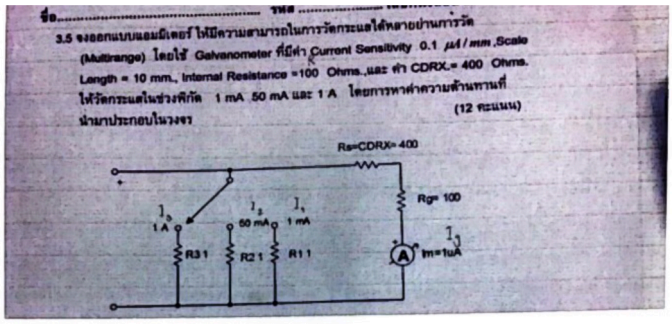
In Ammeters, CDRX requires a SERIES resistor Rs. In Voltmeters, CDRX requires a PARALLEL resistor Rp! Remember this contrast.

CH10-05 | Chapter 10 | Level 2 (Exam / Calc) | Final Exam ข้อ 3.5 (หน้า 29 - 6 คะแนน)

Multi-Range DC Ammeter Design with CDRX

PROBLEM / QUESTION:

Design a 3-range DC Ammeter for ranges I1 = 10 mA, I2 = 50 mA, and I3 = 100 mA using a PMMC meter with Rm = 50 ohms, Ifsd = 1 mA, and CDRX = 450 ohms. Determine the damping resistor Rs and each individual shunt resistor Rsh1, Rsh2, and Rsh3.



รูปประกอบ: Exam Problem 3.5: Multi-Range DC Ammeter with CDRX

KEY FORMULAS & DEFINITIONS:

$$R_s = \text{CDRX} - R_m = 450 - 50 = 400\ \Omega$$

$$R_{\text{branch}} = R_m + R_s = 450\ \Omega$$

$$R_{\text{sh},n} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_n - I_{\text{fsd}}} = \frac{(0.001)(450)}{I_n - 0.001}$$

STEP-BY-STEP MODEL SOLUTION:

1. Determine Series Damping Resistor Rs:
Rs = CDRX - Rm = 450 - 50 = 400 ohms.
Effective meter branch resistance: R_branch = Rm + Rs = 450 ohms.

2. Calculate Shunt Resistors for Each Range:
- Range 1 (10 mA):
Rsh1 = (0.001 * 450) / (0.010 - 0.001) = 0.450 / 0.009 = 50.0 ohms.

- Range 2 (50 mA):
Rsh2 = (0.001 * 450) / (0.050 - 0.001) = 0.450 / 0.049 = 9.184 ohms.

- Range 3 (100 mA):
Rsh3 = (0.001 * 450) / (0.100 - 0.001) = 0.450 / 0.099 = 4.545 ohms.

EXAM PITFALL & CAUTION:

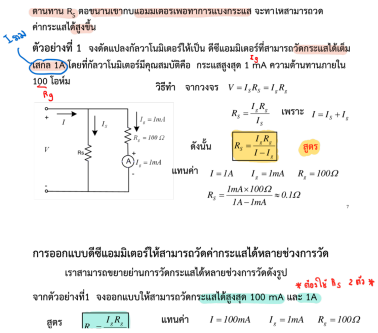
Notice that higher current range requires SMALLER shunt resistance (Rsh3 < Rsh2 < Rsh1) because more current must be bypassed around the meter movement.

CH10-06 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Ayrton Shunt (Universal Shunt) for Multi-Range Ammeter

PROBLEM / QUESTION:

What is an Ayrton Shunt (Universal Shunt)? Why is it superior to individual switched shunts, and how do you calculate the resistor values Ra, Rb, and Rc for multi-range operation?



รูปประกอบ: Ayrton / Universal Shunt Multi-Range Ammeter Circuit

KEY FORMULAS & DEFINITIONS:

$$\text{Total Shunt: } R_{\text{sh}} = R_a + R_b + R_c$$

$$\text{Range 1 (Lowest } I_1\text{): } R_{\text{sh}} = \frac{I_{\text{fsd}} R_m}{I_1 - I_{\text{fsd}}}$$

$$\text{Range 2 (Medium } I_2\text{): } (R_b + R_c) = \frac{I_{\text{fsd}}(R_m + R_a)}{I_2}$$

$$\text{Range 3 (Highest } I_3\text{): } R_c = \frac{I_{\text{fsd}}(R_m + R_a + R_b)}{I_3}$$

STEP-BY-STEP MODEL SOLUTION:

1. What is an Ayrton Shunt:
A tapped chain of resistors (Ra + Rb + Rc) continuously connected across the meter movement. Changing ranges alters which portion acts as shunt and which portion acts as series resistance.

2. Decisive Advantage over Individual Shunts:
- Make-Before-Break Safety: In individual shunts, if the rotary switch momentarily disconnects between contacts, the FULL circuit current would blast through the fragile meter coil (Rm), destroying it immediately.
- Ayrton shunt ensures the meter movement is ALWAYS shunted by resistance, preventing destructive overload during range switching.

EXAM PITFALL & CAUTION:

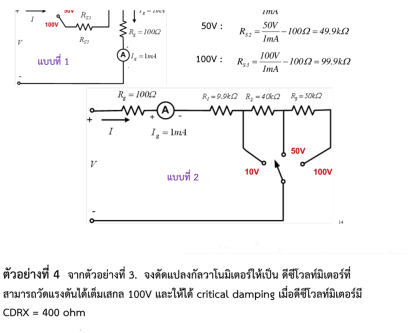
In Ayrton shunt calculations, the resistors not acting as shunt become added in series with Rm! Keep track of which resistors are in parallel vs in series.

CH10-07 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Multi-Range Voltmeter (Individual vs Tapped Series)

PROBLEM / QUESTION:

A meter has Rm = 100 ohms and Ifsd = 50 microamperes. Calculate the multiplier resistor values R1, R2, and R3 to construct a multi-range voltmeter for 10 V, 50 V, and 250 V using a tapped series multiplier configuration.



รูปประกอบ: Tapped Series Multiplier Multi-Range Voltmeter

KEY FORMULAS & DEFINITIONS:

$$S = \frac{1}{I_{\text{fsd}}} = \frac{1}{50 \times 10^{-6}} = 20,000\ \Omega/\text{V} \quad (20\ \text{k}\Omega/\text{V})$$

$$R_{\text{total},1} = S \cdot V_1 = (20\ \text{k}\Omega/\text{V})(10\ \text{V}) = 200\ \text{k}\Omega$$

$$R_1 = R_{\text{total},1} - R_m = 200\ \text{k}\Omega - 100\ \Omega = 199.9\ \text{k}\Omega$$

$$R_2 = S(V_2 - V_1) = 20\ \text{k}\Omega/\text{V} \cdot (50 - 10) = 800\ \text{k}\Omega$$

$$R_3 = S(V_3 - V_2) = 20\ \text{k}\Omega/\text{V} \cdot (250 - 50) = 4,000\ \text{k}\Omega = 4\ \text{M}\Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Voltmeter Sensitivity (S):
S = 1 / Ifsd = 1 / (50 × 10^-6 A) = 20,000 ohms/V (20 k-ohm/V).

2. Range 1 (10 V):
R_total1 = S * 10 V = 200,000 ohms.
R1 = R_total1 - Rm = 200,000 - 100 = 199,900 ohms (199.9 k-ohms).

3. Range 2 (50 V):
R_total2 = S * 50 V = 1,000,000 ohms.
R2 = R_total2 - R_total1 = 1,000,000 - 200,000 = 800,000 ohms (800 k-ohms).

4. Range 3 (250 V):
R_total3 = S * 250 V = 5,000,000 ohms.
R3 = R_total3 - R_total2 = 5,000,000 - 1,000,000 = 4,000,000 ohms (4 M-ohms).

EXAM PITFALL & CAUTION:

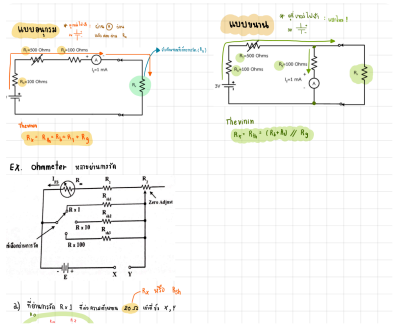
In a tapped series chain, each higher resistor R_n is simply S * (V_n - V_{n-1}), not the full resistance.

CH10-08 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam ข้อ 3

Series-Type Ohmmeter Working Principle & Scale

PROBLEM / QUESTION:

Explain the circuit design of a Series-Type Ohmmeter. Describe its pointer deflection at Rx = 0, Rx = R_half, and Rx = infinity, and explain why its resistance scale is non-linear and reads backwards.



รูปประกอบ: Series-Type Ohmmeter Circuit & Scale

KEY FORMULAS & DEFINITIONS:

$$I = \frac{E}{R_{\text{in}} + R_x}$$

$$R_x = 0 \implies I = I_{\text{fsd}} \quad (\text{Rightmost edge})$$

$$R_x = R_{\text{in}} \implies I = 0.5\, I_{\text{fsd}} \quad (\text{Center of scale: } R_h)$$

$$R_x = \infty \implies I = 0 \quad (\text{Leftmost zero current})$$

STEP-BY-STEP MODEL SOLUTION:

1. Circuit Topology:
A battery E, current-limiting resistor R1, zero-adjust rheostat R2, and meter movement Rm are connected in series with the unknown resistor Rx.

2. Calibration Points:
- Rx = 0 (Short Circuit): Full current flows, pointer deflects to FULL SCALE (far RIGHT) = 0 ohms.
- Rx = R_half (R_in): Total resistance doubles, current drops to 50% Ifsd, pointer sits exactly at MID-SCALE = Rh.
- Rx = infinity (Open Circuit): Current is 0, pointer remains at REST (far LEFT) = infinity ohms.

3. Non-linear & Backwards Scale:
Because current I = E / (R_in + Rx) is an inverse hyperbolic function of Rx, deflection is NOT linear. Scale markings are widely spaced on the right (low ohms) and heavily compressed on the left (high ohms).

EXAM PITFALL & CAUTION:

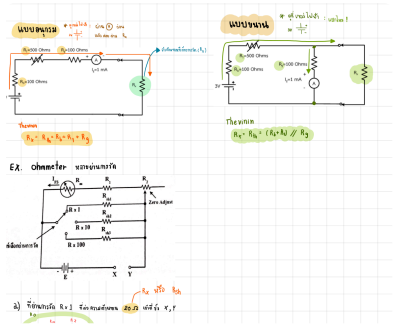
Series ohmmeter scale reads 0 on the RIGHT and infinity on the LEFT (opposite to voltmeters and ammeters).

CH10-09 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Shunt-Type Ohmmeter for Low Resistance Measurement

PROBLEM / QUESTION:

Explain the operating principle of a Shunt-Type Ohmmeter. For what range of resistance is it suited, and how does its scale direction differ from a series ohmmeter?



รูปประกอบ: Shunt-Type Ohmmeter Circuit & Scale

KEY FORMULAS & DEFINITIONS:

$$I_m = I_{\text{total}} \cdot \frac{R_x}{R_m + R_x}$$

$$R_x = 0 \implies I_m = 0 \quad (\text{Leftmost edge} = 0\ \Omega)$$

$$R_x = \infty \implies I_m = I_{\text{fsd}} \quad (\text{Rightmost edge} = \infty\ \Omega)$$

$$\text{Suited for LOW resistance measurement } (5\ \text{m}\Omega - 100\ \Omega)$$

STEP-BY-STEP MODEL SOLUTION:

1. Operating Principle:
The unknown resistor Rx is connected in PARALLEL (shunt) across the meter movement Rm, rather than in series. A series battery and switch provide supply current.

2. Scale Behavior:
- Rx = 0 (Short Circuit): All current bypasses through the zero-ohm short; coil current Im = 0. Pointer stays at far LEFT = 0 ohms.
- Rx = infinity (Open Circuit): No current bypassed; full current flows through Rm, deflecting pointer to far RIGHT = infinity ohms.
- Result: Shunt ohmmeter scale reads from LEFT TO RIGHT (0 on left, infinity on right), exactly matching voltmeter/ammeter scale direction!

3. Primary Application:
Measuring very LOW resistances (0.1 ohm to a few tens of ohms), such as motor winding coils, contact resistance, and transformer wire.

EXAM PITFALL & CAUTION:

Series ohmmeter: 0 is on the RIGHT. Shunt ohmmeter: 0 is on the LEFT. Always know which type the question asks!

CH10-10 | Chapter 10 | Level 1 (Core Concept) | 📌 Lecture Chapter 10 & Lab Multimeter

Multi-Range Ohmmeter & Zero-Adjust Rheostat Function

📌 **PROBLEM / QUESTION:**

What is the primary function of the Zero-Adjust Rheostat (Zero-Ohm Adjust) in an analog multimeter, and how do multi-range multipliers (Rx1, Rx10, Rx10k) scale the center-scale resistance Rh?

📐 **KEY FORMULAS & DEFINITIONS:**

$$R_{\text{scale}} = \text{Reading on Dial} \times \text{Range Multiplier}$$

$$R_{h,\text{new}} = R_h \times \text{Multiplier}$$

$$\text{Battery Aging: } E \downarrow \implies \text{Adjust } R_{\text{zero}} \downarrow \text{ to maintain } I_{\text{fsd}} = \frac{E}{R_{\text{in}}}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Purpose of Zero-Adjust Rheostat:
As the internal dry cell battery ages, its open-circuit terminal voltage E drops and its internal resistance rises. The Zero-Adjust potentiometer compensates for this battery voltage decline by adjusting total internal resistance so that shorting test leads still yields exactly full-scale current Ifsd (pointer aligns perfectly to 0 ohms).

2. Multi-Range Operation:
Switching to higher ranges (e.g., Rx10, Rx1k, Rx10k) changes internal shunt/series divider resistors, multiplying the center-scale half-deflection resistance Rh by that factor. To measure 100 k-ohms accurately, use Rx10k so the pointer lands near center scale where reading precision is highest.

⚠️ **EXAM PITFALL & CAUTION:**

Always re-zero the meter (touch leads together and adjust 0-ohm knob) EVERY TIME you switch resistance ranges!

CH10-11 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Jele Sheet

Rectifier AC Voltmeter & Form Factor

PROBLEM / QUESTION:

A PMMC instrument is used with a full-wave bridge rectifier to measure sinusoidal AC voltage. Derive the relationship between the meter's DC response and the AC RMS voltage, and explain what happens if a square wave or triangular wave is measured.

KEY FORMULAS & DEFINITIONS:

$$V_{\text{avg}} = \frac{2}{\pi} V_m = 0.637\, V_m$$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} = 0.707\, V_m$$

$$\text{Form Factor } k_f = \frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{0.707}{0.637} = 1.11$$

$$V_{\text{scale}} = 1.11 \times V_{\text{avg}}$$

STEP-BY-STEP MODEL SOLUTION:

1. PMMC Mechanism Response:
The PMMC movement responds strictly to average rectified current: V_avg = (2/pi) * Vm = 0.637 Vm.

2. AC Calibration:
Users expect the dial to indicate Effective / Root-Mean-Square (RMS) voltage. For pure sinusoidal AC, the Form Factor is:
kf = V_rms / V_avg = 1.11
Manufacturers calibrate the dial markings by multiplying the actual deflection by 1.11.

3. Non-Sinusoidal Waveform Error:
If measuring non-sinusoidal waveforms (square wave kf = 1.0, triangular wave kf = 1.155), the 1.11 calibration scale causes significant measurement error. True RMS meters (thermal or digital sampling) must be used for non-sine waves.

EXAM PITFALL & CAUTION:

Standard analog AC voltmeters assume a pure sine wave (kf = 1.11). Measuring a 10 V peak square wave will read 11.1 V instead of 10.0 V RMS!

CH10-12 | Chapter 10 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 3.4 (หน้า 30 - 3 คะแนน)

Wattmeter Connection to Load (Current vs Potential Coils)

📌 **PROBLEM / QUESTION:**

Describe the construction of an Electrodynamometer Wattmeter. How are the Current Coil (CC) and Potential/Voltage Coil (PC) connected to a load, and what precautions prevent coil burnout?



รูปประกอบ: Exam Problem 3.4: Electrodynamometer Wattmeter Connection to Load

📐 **KEY FORMULAS & DEFINITIONS:**

$$P = V_{\text{load}} \cdot I_{\text{load}} \cdot \cos\theta$$

$$\text{Current Coil (CC): Fixed, thick wire, low turns} \implies \text{Connected in SERIES with load}$$

$$\text{Potential Coil (PC): Moving, fine wire, high turns} \implies \text{Connected in PARALLEL across load}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Wattmeter Coils & Connection:
- Current Coil (CC / ขดลวดกระแส): Made of thick copper wire with few turns (near-zero impedance). MUST be connected in SERIES with the load to carry full line current.
- Potential Coil (PC / ขดลวดแรงดัน): Made of thin wire with many turns, connected in series with a high multiplier resistance. MUST be connected in PARALLEL across the load terminals to measure line voltage.

2. Power Indication:
Instantaneous torque is proportional to the product i_cc * i_pc, so average pointer deflection indicates real average active power: P = V * I * cos(theta).

3. Burnout Hazard:
NEVER connect the Current Coil in parallel across the voltage supply! Because CC has near-zero resistance, connecting it in parallel creates a dead short circuit, drawing hundreds of amps and instantly vaporizing the coil.

⚠️ **EXAM PITFALL & CAUTION:**

Current Coil connects in SERIES. Potential Coil connects in PARALLEL. Reversing this causes explosive short circuits or an inoperative meter.

CH10-13 | Chapter 10 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 3.1 (หน้า 30 - 2 คะแนน)

Energy Meter (Kilowatt-Hour Meter) Function

📌 **PROBLEM / QUESTION:**

Explain the working principle of an induction-type Kilowatt-Hour Meter (kWh meter). What creates the driving torque on the aluminum disc, and what is the role of the permanent brake magnet?

📐 **KEY FORMULAS & DEFINITIONS:**

$$T_d \propto V \cdot I \cdot \cos\theta = P$$

$$T_b \propto \omega \quad \text{(Eddy-current braking)}$$

$$\text{At steady speed: } T_d = T_b \implies \omega \propto P$$

$$\text{Total Revolutions } N = \int \omega \, dt \propto \int P \, dt = \text{Energy (kWh)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Driving Torque Generation:
The meter contains a shunt electromagnet (connected across line voltage) and a series electromagnet (carrying load current). Their alternating fluxes induce eddy currents inside a lightweight aluminum disc. The interaction between each eddy current and the other magnet's flux creates a continuous rotational driving torque Td proportional to instantaneous real power P.

2. Braking Magnet Role:
A C-shaped permanent magnet sits near the edge of the disc. As the disc spins through its field, eddy currents are induced that create a retarding torque Tb proportional to rotational speed (Tb proportional to omega).

3. Energy Integration:
Equating Td = Tb yields disc speed omega proportional to power P. Therefore, the total number of disc revolutions over time is directly proportional to cumulative electrical energy: Energy = Integral(P dt) in kWh, registered on gear dials.

⚠️ **EXAM PITFALL & CAUTION:**

The disc speed is proportional to POWER (kW), but the total number of revolutions is proportional to ENERGY (kWh).

CH10-14 | Chapter 10 | Level 1 (Core Concept) | Lecture Chapter 10 & Lab Rules

Instrument Accuracy, Precision, and Error Formulas

PROBLEM / QUESTION:

Define the terms Accuracy, Precision, Absolute Error, Relative Error, and Percentage Accuracy in electrical measurement.

KEY FORMULAS & DEFINITIONS:

$$\text{Absolute Error: } e = Y_n - X_n$$

$$\text{Relative Error: } e_r = \frac{Y_n - X_n}{X_n}$$

$$\%\ \text{Error} = \frac{|Y_n - X_n|}{X_n} \times 100\%$$

$$\%\ \text{Accuracy } A = 100\% - \%\ \text{Error}$$

STEP-BY-STEP MODEL SOLUTION:

1. Accuracy (ความแม่นยำ/ความถูกต้อง):
How close the measured value Yn is to the true accepted standard value Xn.

2. Precision (ความเที่ยงตรง):
The repeatability or consistency of successive measurements of the exact same quantity under identical conditions, regardless of whether it is near the true value.

3. Error Formulas:
- Absolute Error: e = Yn - Xn (Measured value minus True value)
- Relative Error: e_r = (Yn - Xn) / Xn
- % Error: (% Error) = (|Yn - Xn| / Xn) * 100%
- % Accuracy: A = 100% - % Error = [1 - |Yn - Xn| / Xn] * 100%.

EXAM PITFALL & CAUTION:

An instrument can have high precision (reads 9.81 V every single time) but low accuracy (if true voltage is 12.00 V).

PART 6

Chapter 11 / Supplement: Semiconductor & Electronic Devices

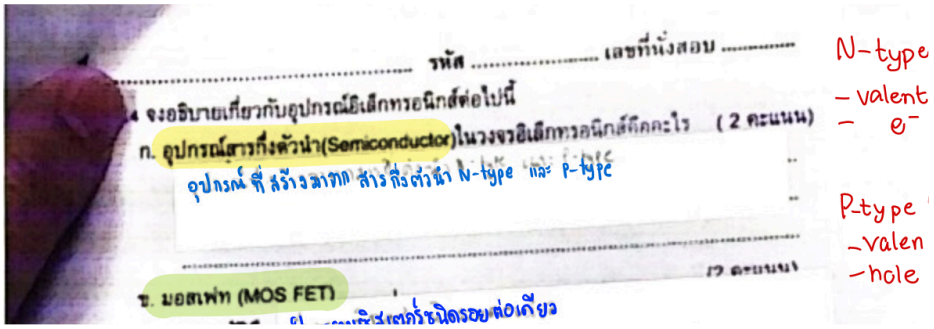
Diodes, Knee Voltages, LEDs, Solar PV Cells, BJT Terminal Currents, MOSFETs, Op-Amps (12 Flashcards)

SEMI-01 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.1 (หน้า 24 - 2 คะแนน)

Intrinsic vs Extrinsic, N-Type vs P-Type Semiconductors

📌 **PROBLEM / QUESTION:**

Distinguish between Intrinsic and Extrinsic semiconductors. Explain how N-Type and P-Type semiconductors are formed by doping, identifying the majority and minority charge carriers in each.



รูปประกอบ: Exam Problem 4.1: Intrinsic & Extrinsic Semiconductors

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{N-Type: Si (4 val)} + \text{Pentavalent (5 val: P, As, Sb)} \implies \text{Majority: Free Electrons } (e^-)$$

$$\text{P-Type: Si (4 val)} + \text{Trivalent (3 val: B, Ga, In)} \implies \text{Majority: Holes } (h^+)$$

$$\text{Mass Action Law: } n \cdot p = n_i^2$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Intrinsic vs Extrinsic Semiconductors:
- Intrinsic: Extremely pure semiconductor crystal (Silicon or Germanium, valence = 4) with no impurities. Electrical conductivity at room temperature is very low.
- Extrinsic: Doped with tiny amounts of impurity atoms to dramatically increase electrical conductivity.

2. N-Type Semiconductor:
- Doped with Pentavalent impurity (valence 5: Phosphorus, Arsenic, Antimony).
- 4 valence electrons form covalent bonds with Silicon, while the 5th electron becomes a free conduction electron.
- Majority Carriers: Free Electrons (e-); Minority Carriers: Holes (h+).

3. P-Type Semiconductor:
- Doped with Trivalent impurity (valence 3: Boron, Gallium, Indium).
- 3 valence electrons bond with Silicon, leaving a vacancy or 'hole' that eagerly accepts an electron.
- Majority Carriers: Holes (h+); Minority Carriers: Electrons (e-).

⚠️ **EXAM PITFALL & CAUTION:**

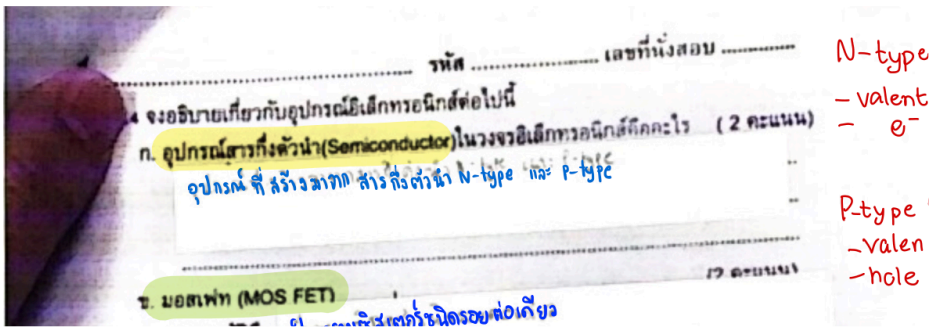
Both N-type and P-type semiconductors remain overall ELECTRICALLY NEUTRAL because total protons equal total electrons in the crystal lattice.

SEMI-02 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.2 (หน้า 24 - 2 คะแนน)

MOSFET Definition, Properties & Uses

📌 **PROBLEM / QUESTION:**

What does MOSFET stand for? Describe its 3-terminal structure, explain why its gate input impedance is near-infinite, and state its primary advantages over BJT transistors.



รูปประกอบ: Exam Problem 4.2: MOSFET Definition & Characteristics

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{MOSFET} = \text{Metal-Oxide-Semiconductor Field-Effect Transistor}$$

$$\text{Terminals: Gate (G), Drain (D), Source (S)}$$

$$I_G \approx 0 \implies R_{\text{in, gate}} \approx 10^{12} - 10^{14}\,\Omega$$

$$I_D = K(V_{GS} - V_{th})^2 \quad \text{(Voltage-Controlled Device)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition & Terminals:
MOSFET stands for Metal-Oxide-Semiconductor Field-Effect Transistor. Its 3 functional terminals are Gate (G), Drain (D), and Source (S) (with a substrate/body terminal internally tied to Source).

2. Near-Infinite Gate Input Impedance:
The Gate electrode is physically isolated from the semiconductor conducting channel by an ultra-thin dielectric layer of Silicon Dioxide (SiO2). Because SiO2 is an exceptional electrical insulator, steady-state gate current IG is practically ZERO (nanoamperes to picoamperes), giving an input impedance of 10^12 to 10^14 ohms.

3. Advantages over BJT:
- Voltage-controlled (requires zero static gate current to remain ON, unlike BJT which demands continuous base current IB).
- Extremely fast switching speeds (ideal for high-frequency switch-mode power supplies and CPU logic gates).
- Low conduction resistance RDS(on) in modern power MOSFETs, yielding high efficiency.

⚠️ **EXAM PITFALL & CAUTION:**

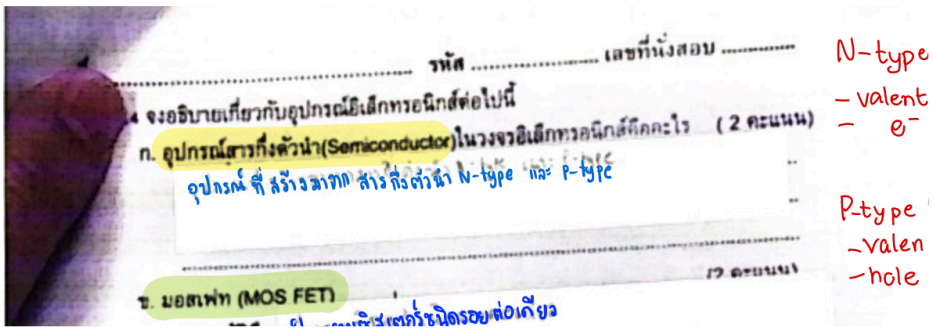
MOSFET is voltage-controlled (Vgs controls Id). BJT is current-controlled (Ib controls Ic).

SEMI-03 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.3 (หน้า 24 - 2 คะแนน)

Integrated Circuit (IC) Definition & Significance

📌 **PROBLEM / QUESTION:**

What is an Integrated Circuit (IC)? How is it manufactured on a single semiconductor wafer, and what revolutionary benefits does it provide over discrete component circuits?



รูปประกอบ: Exam Problem 4.3: Integrated Circuit (IC) Significance

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{IC} = \text{Microscopic integration of Transistors, Diodes, Resistors on a single Silicon chip}$$

$$\text{Moore's Law: Transistor density doubles approximately every 2 years}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition of an IC:
An Integrated Circuit (IC or microchip) is a complete electronic circuit consisting of millions or billions of interconnected microscopic components (transistors, diodes, resistors, capacitors) fabricated monolithically on a single tiny crystal semiconductor substrate (usually Silicon).

2. Revolutionary Advantages over Discrete Circuits:
- Extreme Miniaturization: Packs complex computers into a fingernail-sized silicon chip.
- Mass Production & Low Cost: Fabricated simultaneously in huge batches on silicon wafers using photolithography, drastically lowering cost per gate.
- High Reliability: Monolithic internal metal interconnections eliminate thousands of fragile soldered joints and mechanical wiring failures.
- High Speed & Low Power: Nanometer-scale distances between transistors allow gigahertz switching speeds with negligible parasitic capacitance and minimal power dissipation.

⚠️ **EXAM PITFALL & CAUTION:**

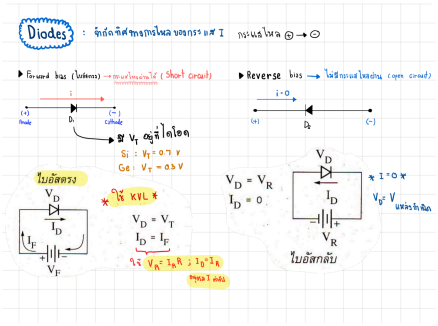
An IC is not merely a single transistor packaged in plastic; it is a COMPLETE SYSTEM of millions of interconnected transistors, diodes, and resistors.

SEMI-04 | Semiconductors | Level 1 (Core Concept) | 📌 Lecture Chapter 10 & Final Exam

PN Junction Diode Biasing & Knee Voltages

📌 PROBLEM / QUESTION:

Explain Forward and Reverse Biasing of a PN junction diode. State the barrier potential (Knee Voltage) for Silicon and Germanium diodes, and explain the mechanism of reverse breakdown.



รูปประกอบ: PN Junction Diode Characteristic Curve & Biasing

📐 KEY FORMULAS & DEFINITIONS:

$$V_{\text{knee, Si}} \approx 0.7\ \text{V} \qquad V_{\text{knee, Ge}} \approx 0.3\ \text{V}$$

$$\text{Forward: } V_{\text{anode}} > V_{\text{cathode}} + V_{\text{knee}} \implies \text{Conducts heavy current } I_D$$

$$\text{Reverse: } V_{\text{cathode}} > V_{\text{anode}} \implies I_D \approx I_s \ (\text{Reverse Saturation Current } \approx \text{nA})$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Forward Bias:
Positive terminal of source connected to Anode (P-type), Negative to Cathode (N-type). When applied voltage exceeds the barrier potential (0.7 V for Si, 0.3 V for Ge), the depletion layer collapses and large forward current flows exponentially.

2. Reverse Bias:
Positive terminal to Cathode (N-type), Negative to Anode (P-type). The external voltage widens the depletion layer, preventing majority carrier flow. Only a tiny temperature-dependent reverse saturation current Is (picoamps to nanoamps) flows.

3. Reverse Breakdown:
If reverse voltage exceeds the Peak Inverse Voltage (PIV / Breakdown Voltage Vz), avalanche multiplication or zener tunneling occurs, causing sudden massive reverse current. Unless current is externally limited by a resistor, the diode will burn out.

⚠️ EXAM PITFALL & CAUTION:

Memorize: Silicon diode Knee Voltage = 0.7 V; Germanium diode Knee Voltage = 0.3 V. Standard problems assume Silicon 0.7 V.

SEMI-05 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Diode Shockley Equation, Dynamic Resistance & LED Circuit

PROBLEM / QUESTION:

State Shockley's ideal diode equation. Define dynamic AC resistance rd, and calculate the current-limiting resistor R required to operate an LED (V_led = 2.0 V, I_led = 15 mA) from a 12 V DC source.

KEY FORMULAS & DEFINITIONS:

$$I_D = I_s\left(e^{\frac{V_D}{nV_T}} - 1\right) \qquad V_T = \frac{kT}{q} \approx 26\ \text{mV at } 300\ \text{K}$$

$$r_d = \frac{nV_T}{I_D} \approx \frac{26\ \text{mV}}{I_D}$$

$$R = \frac{V_{CC} - V_{\text{LED}}}{I_{\text{LED}}} = \frac{12\ \text{V} - 2.0\ \text{V}}{15 \times 10^{-3}\ \text{A}} = 667\ \Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Shockley Diode Equation:
ID = Is * [exp(VD / (n*VT)) - 1]
where Is = reverse saturation current, n = ideality factor (1 to 2), VT = thermal voltage = kT/q = 26 mV at room temperature (27°C / 300 K).

2. Dynamic AC Resistance (rd):
Taking the derivative dVD/dID yields:
rd = 26 mV / ID (where ID is in milliamperes).
At ID = 2 mA, rd = 26 mV / 2 mA = 13 ohms.

3. LED Resistor Calculation:
R = (Vcc - V_led) / I_led = (12 V - 2.0 V) / 0.015 A = 10 V / 0.015 A = 666.67 ohms.
Select standard resistor R = 680 ohms.

EXAM PITFALL & CAUTION:

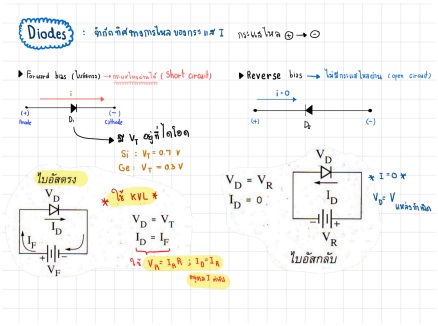
Never connect an LED directly to a voltage source without a current-limiting resistor, as diode current rises exponentially and destroys the LED.

SEMI-06 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam

Zener Diode Voltage Regulator Operation

PROBLEM / QUESTION:

How does a Zener diode function as a voltage regulator? In a circuit with Vin = 15 V to 20 V, Vz = 10 V, Pz_max = 1 W, and load RL = 500 ohms, calculate the required series resistor R.



รูปประกอบ: Zener Diode Voltage Regulator Circuit

KEY FORMULAS & DEFINITIONS:

$$I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{500\ \Omega} = 20\ \text{mA}$$

$$I_{Z,\max} = \frac{P_{Z,\max}}{V_Z} = \frac{1\ \text{W}}{10\ \text{V}} = 100\ \text{mA}$$

$$R_{\min} = \frac{V_{\text{in,max}} - V_Z}{I_{Z,\max} + I_L} = \frac{20 - 10}{0.100 + 0.020} = \frac{10}{0.120} = 83.3\ \Omega$$

$$R_{\max} = \frac{V_{\text{in,min}} - V_Z}{I_{Z,\min} + I_L} = \frac{15 - 10}{0.005 + 0.020} = \frac{5}{0.025} = 200\ \Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Zener Regulation Mechanism:
A Zener diode operates in the REVERSE breakdown region. When reverse-biased beyond its breakdown voltage Vz, terminal voltage remains remarkably constant at Vz over a broad range of Zener current Iz.

2. Load Current:
IL = Vz / RL = 10 V / 500 ohms = 20 mA (constant as long as Vz is regulated).

3. Maximum Zener Current:
Iz_max = Pz_max / Vz = 1 W / 10 V = 100 mA.

4. Calculate Minimum Series Resistance R_min (at Vin_max = 20 V):
R_min = (Vin_max - Vz) / (Iz_max + IL) = (20 - 10) / (0.100 + 0.020) = 10 / 0.120 = 83.3 ohms.

5. Calculate Maximum Series Resistance R_max (at Vin_min = 15 V, assuming Iz_min = 5 mA):
R_max = (Vin_min - Vz) / (Iz_min + IL) = (15 - 10) / (0.005 + 0.020) = 5 / 0.025 = 200 ohms.
Any value between 84 ohms and 200 ohms (e.g. standard 120 ohms) guarantees stable 10 V regulation.

EXAM PITFALL & CAUTION:

Zener diode must be connected in REVERSE BIAS (Cathode to positive supply) to regulate voltage. In forward bias, it behaves like an ordinary 0.7 V diode!

SEMI-07 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 9 & 10 (Renewables)

Solar Photovoltaic Cell Equivalent Circuit & Parameters

PROBLEM / QUESTION:

Draw and explain the electrical equivalent circuit of a Solar Photovoltaic Cell. Define Open-Circuit Voltage (Voc), Short-Circuit Current (Isc), Maximum Power Point (MPP), and Fill Factor (FF).

KEY FORMULAS & DEFINITIONS:

$$I = I_{\text{ph}} - I_D - I_{\text{sh}} = I_{\text{ph}} - I_s\left(e^{\frac{V+IR_s}{nV_T}} - 1\right) - \frac{V + IR_s}{R_{\text{sh}}}$$

$$P_{\max} = V_{\text{mp}} \cdot I_{\text{mp}}$$

$$\text{Fill Factor: } FF = \frac{P_{\max}}{V_{oc} \cdot I_{sc}} = \frac{V_{\text{mp}} \cdot I_{\text{mp}}}{V_{oc} \cdot I_{sc}}$$

$$\text{Efficiency: } \eta = \frac{P_{\max}}{P_{\text{in}}} = \frac{V_{oc} \cdot I_{sc} \cdot FF}{G \cdot A_{\text{cell}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Solar Cell Equivalent Circuit:
Consists of a photocurrent source Iph (proportional to solar irradiance) in parallel with:
- An intrinsic PN junction diode (Shockley equation)
- A shunt resistance Rsh (representing leakage current across cell edges, ideally infinity)
- A series resistance Rs (representing bulk silicon and metal contact resistance, ideally zero).

2. Key Parameters:
- Short-Circuit Current (Isc): Output current when terminals are shorted (V = 0), almost equal to light-generated current Iph.
- Open-Circuit Voltage (Voc): Terminal voltage when net current is zero (I = 0), typically ~0.6 V per silicon cell.
- Maximum Power Point (MPP): The specific point (Vmp, Imp) on the I-V curve where product P = V*I is maximized.
- Fill Factor (FF): Ratio of actual maximum power to theoretical product (Voc * Isc). Higher FF (typically 0.75 - 0.85) denotes a squarer curve and superior cell quality.

EXAM PITFALL & CAUTION:

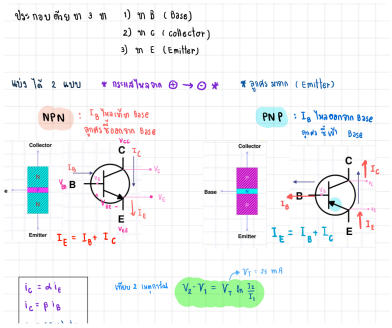
Single silicon solar cell produces only ~0.55 - 0.6 V. That is why 60 or 72 cells are connected in series inside a solar panel to output ~36 - 40 V.

SEMI-08 | Semiconductors | Level 1 (Core Concept) | Lecture Chapter 10 & Homework

BJT Transistor Terminal Currents & Gain Relations

PROBLEM / QUESTION:

State the current relationship between the Emitter (IE), Base (IB), and Collector (IC) of a Bipolar Junction Transistor (BJT). Define DC current gain Beta (hFE) and Alpha, and derive the formula linking Beta and Alpha.



รูปประกอบ: BJT Transistor Terminal Currents & Alpha-Beta Relations

KEY FORMULAS & DEFINITIONS:

$$I_E = I_B + I_C$$

$$\beta = \frac{I_C}{I_B} \qquad \alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{1 - \alpha}$$

STEP-BY-STEP MODEL SOLUTION:

1. Kirchhoff's Current Law for BJT:
By KCL, the total current leaving/entering the emitter equals the sum of base and collector currents:
IE = IB + IC

2. Current Gains Defined:
- Common-Emitter DC Current Gain (Beta or hFE):
Beta = IC / IB (typically ranges from 50 to 300).
- Common-Base DC Current Gain (Alpha):
Alpha = IC / IE (slightly less than 1, typically 0.98 to 0.998).

3. Relationship Derivation:
Divide IE = IB + IC by IC:
IE / IC = (IB / IC) + 1
1 / Alpha = (1 / Beta) + 1 = (Beta + 1) / Beta
Inverting yields: Alpha = Beta / (Beta + 1)
Solving for Beta yields: Beta = Alpha / (1 - Alpha).

EXAM PITFALL & CAUTION:

IE is always the LARGEST current in a BJT: IE = IB + IC. Collector current is slightly smaller than emitter current (Alpha < 1).

SEMI-09 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Lab Experiments

BJT Transistor as an Electronic Switch

PROBLEM / QUESTION:

Explain how an NPN BJT operates as an electronic switch. Define Cutoff and Saturation states, and design a base resistor RB to drive a 12 V, 100 mA relay from a 5 V microcontroller GPIO pin (assume Beta_sat = 10, Vbe_sat = 0.7 V, Vce_sat = 0.2 V).

KEY FORMULAS & DEFINITIONS:

$$\text{Cutoff (OFF)}: V_{in} = 0 \implies I_B = 0,\ I_C = 0,\ V_{CE} = V_{CC}$$

$$\text{Saturation (ON)}: I_B \geq \frac{I_{C,\text{sat}}}{\beta_{\text{sat}}} \implies V_{CE} = V_{CE,\text{sat}} \approx 0.2\ \text{V}$$

$$I_{C,\text{sat}} = \frac{V_{CC} - V_{CE,\text{sat}}}{R_{\text{load}}}$$

$$R_B = \frac{V_{in} - V_{BE,\text{sat}}}{I_B}$$

STEP-BY-STEP MODEL SOLUTION:

1. Operating States as a Switch:
- Cutoff (Switch OFF): When Vin = 0 V, Vbe < 0.7 V. Base current IB = 0, Collector current IC = 0. The transistor behaves as an open switch; full supply Vcc appears across C-E.
- Saturation (Switch ON): Sufficient base current IB is injected so both B-E and B-C junctions become forward-biased. Vce collapses to Vce(sat) ~ 0.2 V. The transistor acts as a closed switch, passing full load current.

2. Base Resistor Calculation:
- Target load current: IC_sat = 100 mA = 0.1 A.
- Overdrive base current for hard saturation (using forced Beta = 10):
IB = IC_sat / Beta_sat = 100 mA / 10 = 10 mA.
- Calculate RB:
RB = (Vin - Vbe_sat) / IB = (5.0 V - 0.7 V) / (10 × 10^-3 A) = 4.3 V / 0.010 A = 430 ohms.
(Select standard 430 ohms or 390 ohms for robust saturation).

EXAM PITFALL & CAUTION:

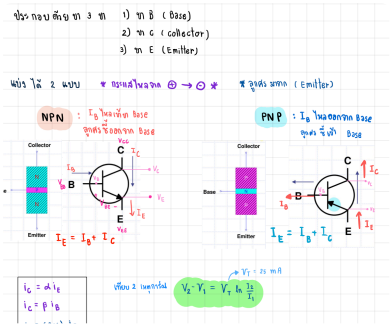
When designing a switching transistor, never use the linear data-sheet Beta (e.g. 100-300). Always use forced Beta_sat = 10 to ensure deep saturation!

SEMI-10 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.2 & Lecture Chapter 10

BJT vs FET / MOSFET Detailed Comparison

📌 PROBLEM / QUESTION:

Provide a comprehensive comparison between Bipolar Junction Transistors (BJT) and Field-Effect Transistors (FET / MOSFET) regarding carrier conduction, control mechanism, input impedance, thermal stability, and noise.



รูปประกอบ: Transistor BJT vs MOSFET Comparison

📐 KEY FORMULAS & DEFINITIONS:

$$\text{BJT: Bipolar (Holes + Electrons) | Current-Controlled } (I_B) | R_{\text{in}} \approx 1 - 10\ \text{k}\Omega$$

$$\text{MOSFET: Unipolar (Majority only) | Voltage-Controlled } (V_{GS}) | R_{\text{in}} \approx 10^{12} - 10^{14}\ \Omega$$

📝 STEP-BY-STEP MODEL SOLUTION:

Comprehensive Comparison:

1. Carrier Conduction:
- BJT is BIPOLAR (current conduction involves both electrons and holes).
- MOSFET is UNIPOLAR (current conduction relies strictly on majority carriers in the channel).

2. Control Mechanism:
- BJT is CURRENT-CONTROLLED: Output current IC is controlled by input base current IB (IC = Beta * IB).
- MOSFET is VOLTAGE-CONTROLLED: Output drain current ID is controlled by gate-to-source voltage VGS.

3. Input Impedance:
- BJT: Low to moderate input impedance (1 k-ohm to 10 k-ohms) because the base-emitter junction is forward-biased.
- MOSFET: Near-infinite input impedance (10^12 to 10^14 ohms) due to the insulating silicon dioxide gate dielectric.

4. Thermal Stability:
- BJT has negative temperature coefficient for Vbe, making it vulnerable to Thermal Runaway.
- MOSFET has positive temperature coefficient for RDS(on), naturally preventing thermal runaway and allowing easy parallel operation.

⚠️ EXAM PITFALL & CAUTION:

BJT = Bipolar, Current-controlled, Low Rin. MOSFET = Unipolar, Voltage-controlled, Extremely high Rin.

SEMI-11 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam

Half-Wave vs Full-Wave Rectifiers & PIV Rating

PROBLEM / QUESTION:

Compare Half-Wave, Full-Wave Center-Tapped, and Full-Wave Bridge Rectifiers regarding DC average output voltage, ripple frequency, efficiency, and Peak Inverse Voltage (PIV) rating across diodes.

KEY FORMULAS & DEFINITIONS:

$$\text{Half-Wave: } V_{\text{dc}} = \frac{V_m}{\pi} = 0.318\,V_m \qquad f_{\text{ripple}} = f_{\text{in}} \qquad \text{PIV} = V_m$$

$$\text{Center-Tapped: } V_{\text{dc}} = \frac{2V_m}{\pi} = 0.637\,V_m \qquad f_{\text{ripple}} = 2f_{\text{in}} \qquad \text{PIV} = 2V_m$$

$$\text{Bridge Rectifier: } V_{\text{dc}} = \frac{2V_m}{\pi} = 0.637\,V_m \qquad f_{\text{ripple}} = 2f_{\text{in}} \qquad \text{PIV} = V_m$$

STEP-BY-STEP MODEL SOLUTION:

1. Half-Wave Rectifier (1 Diode):
- Vdc = Vm / pi = 0.318 Vm
- Ripple frequency = fin (50 Hz)
- Maximum theoretical efficiency = 40.6%
- Diode PIV = Vm

2. Center-Tapped Full-Wave Rectifier (2 Diodes + Center-Tapped Transformer):
- Vdc = 2*Vm / pi = 0.637 Vm
- Ripple frequency = 2*fin (100 Hz, easier to filter)
- Maximum theoretical efficiency = 81.2%
- Diode PIV = 2*Vm (Major drawback: diodes must withstand twice peak voltage!)

3. Bridge Full-Wave Rectifier (4 Diodes, Standard Transformer):
- Vdc = 2*Vm / pi = 0.637 Vm
- Ripple frequency = 2*fin (100 Hz)
- Maximum theoretical efficiency = 81.2%
- Diode PIV = Vm (Only 1*Vm, allowing lower-voltage diodes; dominant industry standard).

EXAM PITFALL & CAUTION:

Crucial exam question: Center-tapped full wave requires PIV = 2*Vm, whereas Bridge rectifier requires only PIV = Vm.

SEMI-12 | Semiconductors | Level 1 (Core Concept) | Lecture Chapter 10 & Lab Rules

Operational Amplifier (Op-Amp 741) Characteristics

PROBLEM / QUESTION:

State the ideal characteristics of an Operational Amplifier (Op-Amp): input impedance, output impedance, open-loop gain, bandwidth, and slew rate. Explain the concept of Virtual Short / Virtual Ground.

KEY FORMULAS & DEFINITIONS:

$$R_{\text{in}} = \infty \implies I_+ = I_- = 0$$

$$R_{\text{out}} = 0 \qquad A_{OL} = \infty \qquad \text{Bandwidth} = \infty$$

$$\text{Virtual Short: } V_+ \approx V_- \quad \text{(Under negative feedback)}$$

$$\text{Inverting Gain: } A_v = -\frac{R_f}{R_{\text{in}}} \qquad \text{Non-inverting Gain: } A_v = 1 + \frac{R_f}{R_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Ideal Op-Amp Characteristics:
- Input Impedance (Rin): Infinite (no current enters the input terminals: I+ = I- = 0).
- Output Impedance (Rout): Zero (can supply any load current without internal voltage drop).
- Open-Loop Voltage Gain (A_OL): Infinite.
- Bandwidth (BW): Infinite (amplifies signals from DC to infinite frequency without attenuation).
- Slew Rate: Infinite (output changes voltage instantaneously).
- Common-Mode Rejection Ratio (CMRR): Infinite.

2. Virtual Short & Virtual Ground:
Because open-loop gain A_OL is infinite, in any negative feedback configuration, the differential input voltage MUST be zero:
(V+ - V-) = Vout / A_OL = Vout / infinity = 0 ⇒ V+ = V-.
This is called a 'Virtual Short'. If the non-inverting terminal V+ is tied to physical ground (0 V), then V- is held firmly at 0 V without being physically connected to ground, known as 'Virtual Ground'.

EXAM PITFALL & CAUTION:

Virtual short (V+ = V-) holds ONLY when negative feedback is present. In an open-loop comparator, V+ is NOT equal to V-.